Fall 2025: Homotopical Methods in Algebraic Topology

Yutong Chen *

Contents

1 Introduction **2**

2 Simplicial Set **3**

3 Homotopy Groups **12**
3.1 Definition of Homotopy Groups . . . 12
3.2 Long Exact Sequence of Homotopy Groups . . . 13
3.3 Whitehead Theorem . . . 14

4 ∞-category of Spaces **17**
4.1 First Construction . . . 17
4.2 Second Construction . . . 18
4.3 Third Construction . . . 19
4.4 Homotopy Pullback . . . 19

5 Simplicial R-modules **22**

6 Spectral Sequence **24**
6.1 Homological Spectral Sequence . . . 24
6.2 Construction of Serre Spectral Sequence . . . 27
6.3 Cohomological Spectral Sequence . . . 28
6.4 Computation of Homotopy Group . . . 31
6.5 Whitehead Tower . . . 33
6.6 Mod $\mathcal{C}$ Theory . . . 35

7 Homotopy Group of Spheres **37**
7.1 Homotopy Pushout . . . 37
7.2 Recent Progress on Stable Homotopy Groups . . . 39

8 Cohomology of Eilenberg-Maclane Spaces **40**

*based on lectures by Jeremy Hahn

9 Introduction to Spectra **43**
9.1 $\mathbb{E}_1$-Spaces . . . 43
9.2 $\mathbb{E}_\infty$-Spaces . . . 44
9.3 Spectra . . . 46
9.4 Cohomology and Homology . . . 49
9.5 Spectral Sequence for Spectra . . . 52

10 Higher Algebra **53**
10.1 Homotopy Ring . . . 53
10.2 Modules . . . 56
10.3 Dual Steenrod Algebra . . . 61
10.4 A Study on MU . . . 63
10.5 More on Formal Group Laws . . . 68

1 Introduction

Recall every category $\mathcal{C}$ has a core groupoid $\mathrm{Core}(\mathcal{C})$. Given a category $\mathcal{C}$, a basic question is to classify objects of $\mathcal{C}$ up to isomorphism.

Example. *Let $\mathcal{C} = \mathsf{Set}$, isomorphisms are bijections. Objects up to isomorphisms are classified up to cardinalities.*

Example. *For every commutative ring R, $\mathcal{C} = R\text{-}\mathsf{mod}$. When R is a field, objects are classified by dimension.*

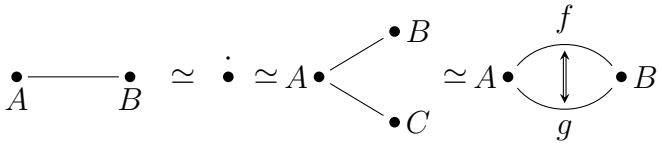
Example. *Let $\mathcal{C} = \mathsf{Top}$, the category of (nice) topological spaces. We might try to classify objects up to isomorphism. A useful tool is $H_i(-, R) : \mathsf{Top} \to R\text{-}\mathsf{mod}$. A key ingredient in the definition of these functors is the study of maps $|\Delta^n| \to X$*

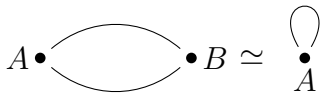
Question: How to think about objects in $\mathsf{Ho}(\mathsf{Top})$? Roughly, a homotopy type consists of

- A collection of objects
- For every pair of objects, a collection of isomorphisms
- For every pair $f, g : A \to B$ of parallel isomorphisms, there is a collection of 2-isomorphisms $f \Rightarrow g$.
- $\cdots$

Example. *A set is a homotopy type with no nonidentity isomorphisms, 2 isomorphisms, etc. A groupoid is a homotopy type with no nonidentity 2-isomorphisms, 3-isomorphisms, etc.*

Example. *1.*



2. 

Remark. $\mathsf{Ho}(\mathsf{Top})$ *is sometimes called the category of* ∞*-groupoids.*

Example. *Every space has an underlying* ∞*-groupoid.*

- *Objects are points in* X *(functions* $|\Delta^0| \to X$*)*
- *isomorphisms are paths in* X *(functions* $|\Delta^1| \to X$*)*
- *A map* $|\Delta^2| \to X$ *is the data of a 2-isomorphism* $g \circ f \Rightarrow h$

2 Simplicial Set

Definition 2.1. For each integer $n \geq 0$, there is a category $[n]$: objects of $[n]$ are $\{0, \cdots, n\}$,

$$\mathrm{Hom}_{[n]}(i,j) = \begin{cases} \emptyset, & \text{if } i > j \\ *, & \text{if } i \leq j \end{cases}$$

The simplex category $\mathbf{\Delta}$ has one object $[n]$ for every integer $n \geq 0$, and $\mathrm{Hom}_{\mathbf{\Delta}}([n],[m])$ is the set of functors from $[n]$ to $[m]$ = the set of order preserving functions from $\{0, \cdots, n\}$ to $\{0, \cdots m\}$. In particular $\mathbf{\Delta} \hookrightarrow \mathsf{Cat}$ is fully faitful.

Example. *There are 6 functors from* $[1]$ *to* $[2]$*.*

Example. *There is a unique morphism from* $[n]$ *to* $[0]$*.*

Remark. *There is a functor* $\mathbf{\Delta} \to \mathsf{Top}$*,* $[n] \to |\Delta^n|$ *and the morphisms are sent to corresponding affine extensions on the level of standard simplexes.*

Definition 2.2. A simplicial set is a functor $\mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Set}$

Example. *Let* X *be a topological space. There is a simplicial set*

$$\mathrm{Sing}(X) : \mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Set}$$

$$[n] \mapsto \mathrm{Hom}_{\mathsf{Top}}(|\Delta^n|, X)$$

Note $H_i(X,R)$ *depends only on* $\mathrm{Sing}(X)$*.*

Remark. *Not every simplicial set arises as Sing of a space. In fact,* $\mathrm{Sing}(X)$ *of a topological space have a few special properties that we will discuss later.*

Example. *Let* $\mathcal{C}$ *be a category. There is a simplicial set "nerve of the category":*

$$N(\mathcal{C}) : \mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Set}$$

$$[n] \mapsto \{ \text{ set of functors } [n] \to \mathcal{C} \}$$

In fact, we will later see that categories are completely determined by their nerves.

There are names for certain morphisms in $\mathbf{\Delta}$: $\delta^i : [n-1] \to [n]$ the order preserving injection that omits i. These are called face maps. If apply $\mathbf{\Delta} \to \mathsf{Top}$, then δ^i is sent to the inclusion of a face.

There are also degeneracy maps: $\sigma^i : [n+1] \to [n]$ the order preserving surjecting repeating i.

Remark. *Every morphism in $\mathbf{\Delta}$ is a composite of face and degeneracy maps.*

Explicitly, a simplicial set $X : \mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Set}$ is a sequence of sets $X_0, X_1, \cdots$ with $X_i = X([i])$ with maps of sets

$$d_0, \cdots, d_n : X_n \to X_{n-1}$$

induced by δ^i and

$$s_0, \cdots, s_n : X_n \to X_{n+1}$$

induced by σ^i subject to certain relations. We call X_i the set of i-simplices.

Remark. *A morphism of simplicial sets is a natural transformation of functors $\mathbf{\Delta}^{op} \to \mathsf{Set}$. This can be spelled out by giving maps $X_i \to Y_i$ making all possible diagrams commute. Hence there is a category of simplicial sets sSet. So we have functors:* $\mathrm{Sing} : \mathsf{Top} \to \mathsf{sSet}$ *and* $N : \mathsf{Cat} \to \mathsf{sSet}$.

Example. *If $\mathcal{C}$ is a category. Recall $N(\mathcal{C})_n = \mathrm{Hom}_{\mathsf{Cat}}([n], \mathcal{C})$. More concretely:*

$$N(\mathcal{C})_0 = \mathrm{Fun}([0], \mathcal{C}) = \text{set of objects of } \mathcal{C}$$

$$N(\mathcal{C})_1 = \mathrm{Fun}([1], \mathcal{C}) = \text{set of morphisms of } \mathcal{C} = \coprod_{c, c' \in \mathcal{C}} \mathrm{Hom}_{\mathcal{C}}(c, c')$$

$N(\mathcal{C})_2 = \mathrm{Fun}([2], \mathcal{C}) =$ set of pair of composable arrows which determine their composite

s_0 picks out identity morphisms, d_0 picks out the target of a morphism, d_1 picks out the source, etc.

Remark. sSet *is a presheaf category.*

If $\mathcal{C}$ is any small category, then $\mathsf{PSh}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathsf{Set})$. For presheaf category, we have the Yoneda embedding:

$$Y : \mathcal{C} \to \mathsf{PSh}(\mathcal{C})$$

$$c \mapsto \mathrm{Hom}_{\mathcal{C}}(-, c)$$

The Yoneda lemma a says this is fully faithful. Presheaves in the image of Y are called representable.

Definition 2.3. $\Delta^n = Y([n]) \in \mathsf{sSet}$.

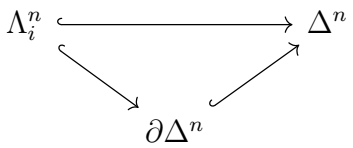
Yoneda lemma tells us that if $X \in \mathsf{sSet}$, then $\mathrm{Hom}_{\mathsf{sSet}}(\Delta^n, X) = X_n$.

Explicitly, $(\Delta^n)_m$ consists of order preserving functions $[m] \to [n]$. For example, $(\Delta^2)_0$ has three elements, $(\Delta^2)_1$ has six elements (three of which are degenerate, i.e. in the image of one of the s_i maps).

Notations: Let $(\partial\Delta^n)_m$ be the order preserving maps from $[m]$ to $[n]$ that are not surjective. There is a natural inclusion $\partial\Delta^n \to \Delta^n$.

For $0 \leq i \leq n$, there is an inclusion $\Lambda^n_i \to \Delta^n$, the "i-th horn" of Δ^n. It consists of order preserving maps with image not containing all of $0, 1, \cdots, i-1, i+1, \cdots, n$.

In particular, we have a factorization:



Definition 2.4. An inner horn inclusion is one of the maps $\Lambda^n \to \Delta^n$ for $0 < i < n$. An outer horn inclusion is either $\Lambda^n_0 \to \Delta^n$ or $\Lambda^n_n \to \Delta^n$.

Proposition 2.5. *Let $\mathcal{C}$ be a category. Then any map $\Lambda^n_i \to N(\mathcal{C})$ for $0 < i < n$ extends uniquely to a map $\Delta^n \to N(\mathcal{C})$.*

Remark. *In fact, a simplicial set is the nerve of a category iff the proposition holds.*

Remark. *If $\mathcal{C}$ and $\mathcal{D}$ are categories, then the set of functors from $\mathcal{C}$ to $\mathcal{D}$ is in bijection with the set* $\mathrm{Hom}_{\mathsf{sSet}}(N(\mathcal{C}), N(\mathcal{D}))$.

Proposition 2.6. *A category is a groupoid if and only if it also has unique outer horn fillings.*

Proposition 2.7. *There are adjoint functors*

$$|\quad|\colon \mathsf{sSet} \to \mathsf{Top}$$

called "geometric realizations" the left adjoint, and

$$\mathrm{Sing} : \mathsf{Top} \to \mathsf{sSet}$$

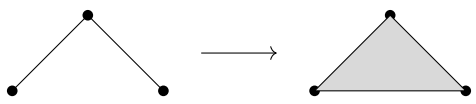
the right adjoint. This is characterized by Δ^n is the topological n-simplex.

Theorem 2.8. *If $\mathcal{C}$ is a small category, then $\mathsf{PSh}(\mathcal{C})$ has all colimits and limits. For any functor $F : \mathcal{C} \to \mathcal{D}$, to a category $\mathcal{D}$ with colimits, there is a unique colimit preserving functor $\mathsf{PSh}(\mathcal{C}) \to \mathcal{D}$ such that the functor*

$$\mathcal{C} \xrightarrow{Y} \mathsf{PSh}(\mathcal{C}) \to \mathcal{D}$$

is F. This colimit preserving functor is left adjoint.

Example. *Consider the inclusion $\Lambda^2_1 \hookrightarrow \Delta^2$. The induced map on the geometric realizations is precisely:*

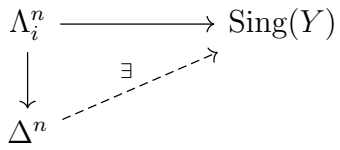


Now if $Y \in \mathsf{Top}$, then what is $\mathrm{Sing}(Y)_n$? In fact, we have:

$$\mathrm{Sing}(Y)_n = \mathrm{Hom}_{\mathsf{sSet}}(\Delta^n, \mathrm{Sing}(Y)) = \mathrm{Hom}_{\mathsf{Top}}(|\Delta^n|, Y)$$

by adjunction.

Proposition 2.9. *If Y is a topological space, for any map $\Lambda^n_i \to \mathrm{Sing}(Y)$ admits a lift to an n-simplex*



Proof. By adjunction, this is equivalent to producing a diagram on the level of geometric realization. But note $|\Lambda^n_i| \to |\Delta^n|$ in fact admits a retract. □

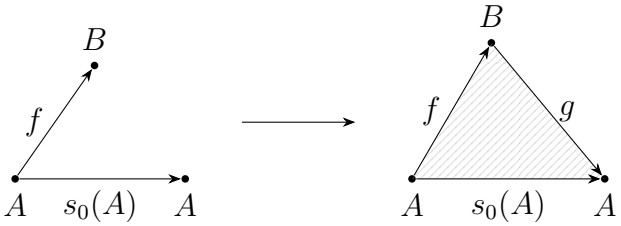
Definition 2.10. A *Kan complex* (or ∞- *groupoid*) is a simplicial set X that admits a lift for every horn filler.

Example. $\mathrm{Sing}(Y)$ *is a Kan complex.*

Example. *Let $\mathcal{C}$ be a category. It's a groupoid iff it's a Kan complex*

Definition 2.11. A *quasicategory*(or ∞*-category*) is a simplicial set with (not necessarily unique) inner horn fillers.

Remark. *Suppose $f : A \to B$ is a morphism in a Kan complex $\mathcal{C}$. Then there exists an "inverse morphism" $g : B \to A$:*



Suppose $\mathcal{C}$ is a quasicategory. Then $\mathcal{C}_0$ are called objects of $\mathcal{C}$; $\mathcal{C}_1$ are morphisms in $\mathcal{C}$; For $E \in \mathcal{C}_2$, we say $d_1 E \in \mathcal{C}$ is 2-isomorphic to the composite of $d_0 E, d_2 E \in \mathcal{C}$.

Definition 2.12. We say two maps $f, g : X \to Y$ of Kan complexes are homotopic if there exists $h : X \times \Delta^1 \to Y$ of simplicial sets such that:

$$X \times \Delta^0 \xrightarrow{1_X \times d_0} X \times \Delta^1 \hookrightarrow Y \text{ is } g$$

$$X \times \Delta^0 \xrightarrow{1_X \times d_1} X \times \Delta^1 \hookrightarrow Y \text{ is } f$$

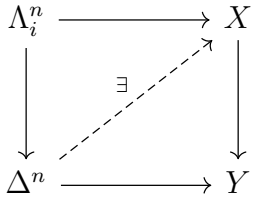
This is an equivalence relation, and using it we can define what it means for two Kan complexes to be homotopy equivalent.

Theorem 2.13 (Quillen)**.** *If X is a nice topological space (e.g. CW complex), then* $|\mathrm{Sing}(X)| \simeq X$*. Also, if X is a Kan complex,* $\mathrm{Sing}(|X|) \simeq X$*. Therefore:*

$$\mathsf{Ho(Top)} \simeq \mathsf{Ho(Kan)}$$

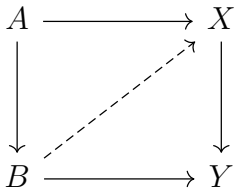
is an equivalence of categories.

Definition 2.14. A morphsim $X \to Y$ of simplicial sets is a Kan fibration if it admists horn fillers



Remark. *A simplicial set X is a Kan complex iff the unique map $X \to \Delta^0$ is a Kan fibration.*

Question: For what maps $A \to B$ of simplicial sets is it the case that Kan fibrations admits lifts against $A \to B$



Definition 2.15. A class Σ of morphisms of simplicial sets is said to be saturated if it is closed under pushouts, retracts, coproducts, composition, and transifinite composition.

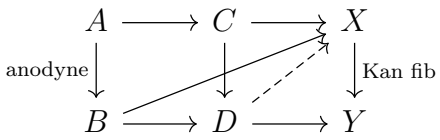
Definition 2.16. The smallest sturated class of maps containing all horn inclusions is called the class of *anodyne maps*.

Remark. *The answer to the question will be $A \to B$ anodyne.*

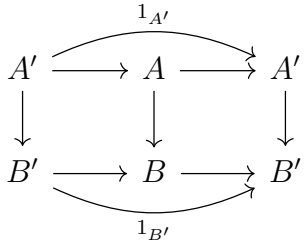
Remark. *A morphism of simplicial sets $A \to B$ is anodyne if and only if:*

1. *a monomorphism*
2. *$|A| \to |B|$ is a homotopy equivalence.*

If $A \to B$ is an anodyne map, and $A \to C$ is any map, then $C \to D = B \coprod_A C$ is anaodyne.



If $f : A \to B$ is anodyne, so is any $g : A' \to B'$ that fits in a diagram



so that g is a retract of of f.

If $\{f_i : A_i \to B_i\}_{i \in I}$ is a collection of anodyne maps, then

$$\coprod_{i \in I} f_i : \coprod_{i \in I} A_i \to \coprod_{i \in I} B_i$$

is anodyne.

If $A \to B$, $B \to C$ are anodyne, so is their composit (easy to check).

Suppose given a diagram of simplicial sets

$$A_0 \xrightarrow{f_0} A_1 \xrightarrow{f_1} A_2 \xrightarrow{f_2} A_3 \xrightarrow{f_3} \cdots$$

such that each f_i is anodyne. Then the map $A_0 \to \varinjlim A_i$ is anodyne.

Goal: We will prove that if $f : A \to B$ is anodyne, and C is a simplicial set, then $A \times C \xrightarrow{f \times 1_C} B \times C$ is anodyne.

Remark. *All anodyne maps are monomorphisms.*

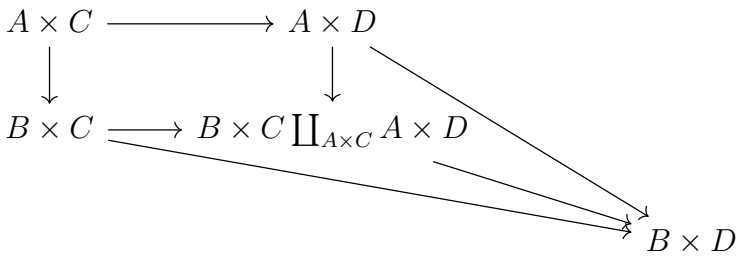
Definition 2.17. Given maps $A \to B$ and $C \to D$ of simplicial sets. The pushout product

$$(A \to B) \boxtimes (C \to D)$$

is the map

$$B \times C \coprod_{A \times C} A \times D \to B \times D$$

given by the following diagram:



Remark. *Consider $(A \to B) \boxtimes (\emptyset \to C)$. This is $A \times C \to B \times C$*

Theorem 2.18. *Suppose $A \to B$ is monomorphism of simplicial sets, and $C \to D$ is anodyne. Then $(A \to B) \boxtimes (C \to D)$ is anodyne.*

Remark. *Pushout products are associative:* $i \boxtimes (j \boxtimes k) = (i \boxtimes j) \boxtimes k$.

Lemma 2.19. *The class of anodyne map is the smallest saturated class containing all maps of the form*

$$(A \times B) \boxtimes (\Delta^0 \to \Delta^1)$$

where $A \to B$ *is a monomorphism.*

Proof. (of theorem given the lemma) Fix a monomorphism i. Consider the class of maps j such that $i \boxtimes j$ is anodyne. It's easy to see this class is saturated. Thus it suffices to show that

$$i \boxtimes ((A \to B) \boxtimes (\Delta^0 \to \Delta^1))$$

is anodyne for all monomorphism $A \to B$. But this follows immediately from the associativity of pushout products and the fact that pushout products preserve monomorphisms. □

Proof. (of lemma)

1. Every horn inclusion is in the saturated class generated by $(A \to B) \boxtimes (\Delta^0 \to \Delta^1)$

   Consider a horn inclusion $\Lambda^n_i \to \Delta^n$ for $i < n$, we will produce a diagram

$$\begin{array}{ccccc} \Lambda^n_i & \longrightarrow & (\Lambda^n_i \to \Delta^n) \boxtimes (\Delta^0 \to \Delta^1) & \longrightarrow & \Lambda^n_i \\ \downarrow & & \downarrow & & \downarrow \\ \Delta^n & \longrightarrow & \Delta^n \times \Delta^1 & \longrightarrow & \Delta^n \end{array}$$

   which is a retract. The data of the diagram is contained in the bottom row. THe first bottom map is $\Delta^n \times \{1\} \to \Delta^n \times \Delta^1$, the second map is depicted as:

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & 1 & \longrightarrow & 2 & \longrightarrow \cdots \longrightarrow & i-1 & \longrightarrow & i & \longrightarrow \cdots \longrightarrow & i \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & 1 & \longrightarrow & 2 & \longrightarrow \cdots \longrightarrow & i-1 & \longrightarrow & i & \longrightarrow \cdots \longrightarrow & n \end{array}$$

2. Every map of the form $(A \to B) \boxtimes (\Delta^0 \to \Delta^1)$ is anodyne. We need the following:

   Proposition. *Suppose* $A \to B$ *is a monomorphism of simplicial sets. Then there is a sequence of monomorphisms*

$$A \to A_0 \to A_1 \to A_2 \cdots \to B$$

   such that $B = \operatorname{colim}(A \to A_0 \to A_1 \to \cdots)$ *and there exist pushout squares:*

$$\begin{array}{ccc} \coprod \partial\Delta^i & \longrightarrow & A_{i-1} \\ \downarrow & & \downarrow \\ \coprod \Delta^i & \longrightarrow & A_i \end{array}$$

   Here A_i *is the smallest simplicial subset of* B *containing* A *such that* $(A_i)_j \to B_j$ *is an isomorphism for* $j \leq i$.

So we are reduced to proving that

$$(\partial\Delta^n \to \Delta^n) \boxtimes (\Delta^0 \to \Delta^1)$$

is anodyne for every $n \geq 0$.

Example. *The case for $n = 1$ is the morphism $(\partial\Delta^1 \to \Delta^1) \boxtimes (\Delta^0 \to \Delta^1)$. This is can be done by pushing out against two horn inclusions.*

□

Suppose X, Y are simplicial sets. Define $\underline{\mathrm{Hom}}_{\mathsf{sSet}}(X, Y)$ to be the simplicial set with n-simplices $\mathrm{Hom}_{\mathsf{sSet}}(X \times \Delta^n, Y) = \mathrm{Hom}_{\mathsf{sSet}}(\Delta^n \times X, Y) = \mathrm{Hom}_{\mathsf{sSet}}(\Delta^n, \underline{\mathrm{Hom}}(X, Y))$.

Remark. *Simplicial set is a cartesian closed category, namely:*

$$\mathrm{Hom}_{\mathsf{sSet}}(A, \underline{\mathrm{Hom}}(B, C)) \simeq \mathrm{Hom}_{\mathsf{sSet}}(A \times B, C)$$

Proposition 2.20. *Suppose X, Y are simplicial sets and Y is a Kan complex. Then $\underline{\mathrm{Hom}}(X, Y)$ is a Kan complex.*

Proof.

$$\begin{array}{ccc} \Lambda_i^n & \longrightarrow & \underline{\mathrm{Hom}}(X, Y) \\ \downarrow & \nearrow & \\ \Delta^n & & \end{array} \iff \begin{array}{ccc} \Lambda_i^n \times X & \longrightarrow & Y \\ \downarrow & \nearrow & \\ \Delta^n \times X & & \end{array}$$

Now f is $(\Lambda_i^n \to \Delta^n) \boxtimes (\emptyset \to X)$, so it is anodyne. □

Definition 2.21. Two objects in a Kan complex are isomorphism if there exists a map between them.

Definition 2.22. We say that a map $f : X \to Y$ of Kan complexes is a homotopy equivalence if there exists $g : Y \to X$ such that $f \circ g$ is isomorphic to 1_Y in $\underline{\mathrm{Hom}}(Y, Y)$ and $g \circ f$ is isomorphic to 1_X in $\underline{\mathrm{Hom}}(X, X)$.

Definition 2.23. If X is a Kan complex, $\pi_0 X$ is the set of isomorphism classes of objects in X.

Definition 2.24. The saturation of inner horn inclusions is called the class of inner anodyne maps. An inner fibration is a map with lifts against inner anodyne maps. A simplicial set is a quasicategory if $X \to \Delta^0$ is an inner fibration.

Remark. *Not every anodyne map is inner anodyne.*

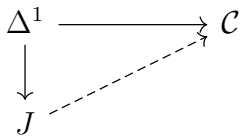
Example. *Let J be the universal isomorphism. Then $\Delta^1 \to J$ is anodyne but not inner anodyne.*

Remark. *The saturation of boundary maps $\{\partial\Delta^n \to \Delta^n : n \geq 0\}$ is precisely the monomorphisms. Because any monomorphism is canonically a transfinite composition of cell attachment. A map with a lift against all monomorphisms is a trivial fibration.*

Theorem 2.25. *If i is anodyne (resp. inner anodyne) and j is a monomorphism, then $i \boxtimes j$ is anodyne (resp. inner anodyne).*

Corollary 2.26. *If X is a simplicial set, Y is a Kan complex, then $\underline{\mathrm{Hom}}(X, Y)$ is a Kan complex. If X is a simplicial set, Y is a quasicategory, then $\underline{\mathrm{Hom}}(X, Y)$ is a quasicategory.*

Definition 2.27. If $\mathcal{C}$ is a quasicategory, a morphism $\Delta^1 \to \mathcal{C}$ is called an isomorphism if it extends to a diagram:

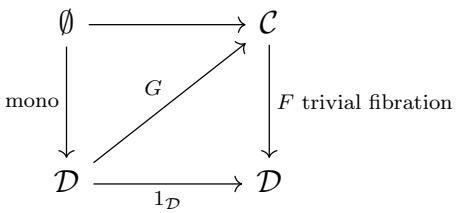


Definition 2.28. Let $\mathcal{C}, \mathcal{D}$ be ∞-categories. A *functor* between $\mathcal{C}$ and $\mathcal{D}$ is a morphism between the corresponding simplicial sets. A *natural transformation* is a morphism in $\underline{\mathrm{Hom}}(\mathcal{C}, \mathcal{D})$. A *natural isomorphism* is a functor $J \to \underline{\mathrm{Hom}}(\mathcal{C}, \mathcal{D})$.

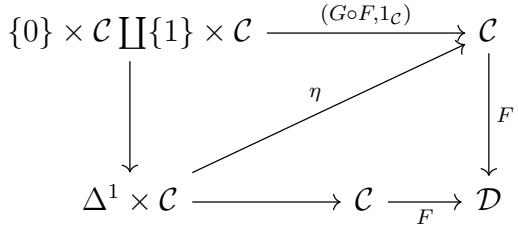
A functor $F : \mathcal{C} \to \mathcal{D}$ is an *equivalence* of quasicategories if there exists $G : \mathcal{D} \to \mathcal{C}$ such that $F \circ G$ isomorphic to $1_{\mathcal{D}} \in \underline{\mathrm{Hom}}(\mathcal{D}, \mathcal{D})$, and $G \circ C$ isomorphic to $1_{\mathcal{C}}$. In the special case where $\mathcal{C}$ and $\mathcal{D}$ are Kan complexes, equivalences are called *homotopy equivalence*.

Theorem 2.29. *A trivial fibration between quasicategories is an equivalence.*

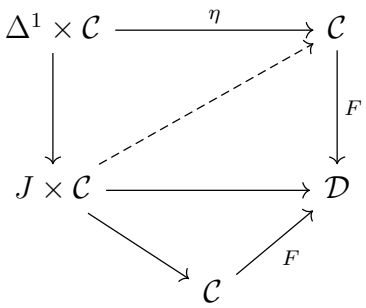
Proof. Suppose $F : \mathcal{C} \to \mathcal{D}$ is a trivial fibration. The following diagram tells us we can construct $G : \mathcal{D} \to \mathcal{C}$ such that $F \circ G = 1_{\mathcal{D}}$.



It remains to show $G \circ F \simeq 1_{\mathcal{C}}$. The following diagram enables us to construct a natural transformation η between them:



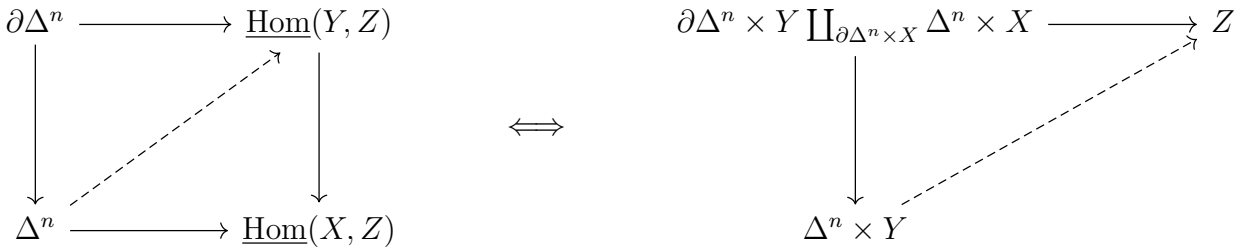
Finally we should check η is a natural isomorphism:



□

Theorem 2.30. *Suppose $f : X \to Y$ is an anodyne map between Kan complexes. Then f is an equivalence.*

Proof. I claim $\underline{\mathrm{Hom}}(Y, Z) \to \underline{\mathrm{Hom}}(X, Z)$ is a trivial fibration for every Kan complex Z. Indeed we need to provide a lift:



But the later is a pushout product of an anodyne map against a monomorphism. In particular

$$\underline{\mathrm{Hom}}(Y, Y) \to \underline{\mathrm{Hom}}(X, Y) \quad \underline{\mathrm{Hom}}(Y, X) \to \underline{\mathrm{Hom}}(X, X)$$

are trivial fibrations, i.e. equivalences of Kan complexes. □

3 Homotopy Groups

3.1 Definition of Homotopy Groups

Recall if $X \in \mathsf{Top}, x \in X$ a point. We have $\pi_1(X, x)$.

Definition 3.1. Suppose $A \subset B$ and $X \subset Y$ are monomorphisms of Kan complexes. Then $\underline{\mathrm{Hom}}((B, A), (Y, X))$ is the pullback

$$\begin{array}{ccc} & & \underline{\mathrm{Hom}}(B, Y) \\ & & \downarrow \\ \underline{\mathrm{Hom}}(A, X) & \longrightarrow & \underline{\mathrm{Hom}}(A, Y) \end{array}$$

This is the simplicial set of maps from B to Y with the property that A maps inside of X. This is a Kan complex.

Definition 3.2. Suppose X is a Kan complex and $x \in X_0$ is a point/object in X. Then we define:

$$\pi_1(X, x) = \pi_0 \underline{\mathrm{Hom}}((\Delta^1, \{0\} \coprod \{1\}), (X, \{x\}))$$

Definition 3.3. We use $\square^n$ to denote $(\Delta^1)^{\times n}$, and $\partial\square^n$ to denote the simplicial set

$$\bigcup_j (\Delta^1)^{\times j} \times (\{0\} \coprod \{1\}) \times (\Delta^1)^{n-j-1}$$

Definition 3.4. Let X be a Kan complex, and $x \in X_0$ is a point. We define:

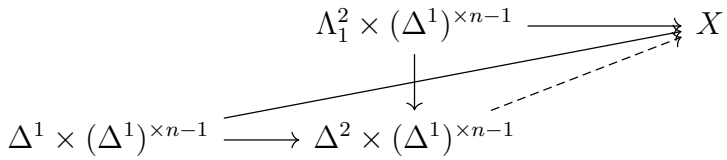
$$\pi_n(X, x) = \pi_0 \underline{\mathrm{Hom}}((\square^n, \partial\square^n), (X, \{x\}))$$

for any $n \geq 1$.

Remark. *There is a unital multiplication*

$$\pi_n(X, x) \times \pi_n(X, x) \to \pi_n(X, x)$$

Given $f : \square^n \to X$, $g : \square^n \to X$, we can form



There are in fact n different unital multiplications on π_n: $\cdot_1, \cdot_2, \cdots, \cdot_n$.

Theorem 3.5 (Eckman-Hilton)**.** *Suppose A is a set with two unital multiplications $(\cdot_1, u_1)$ and $(\cdot_2, u_2)$. Suppose $(a \cdot_1 b) \cdot_2 (c \cdot_1 d) = (a \cdot_2 c) \cdot_1 (b \cdot_2 d)$. Then $\cdot_1 = \cdot_2$, $u_1 = u_2$ and $\cdot_1 = \cdot_2$ is associative and commutative.*

Corollary 3.6. *These multiplications are all equal and associative, and for $n \geq 2$ commutative. In particular, $\pi_n(X, x)$ is an abelian group for $n \geq 2$.*

Why care to calculate homotopy groups? Many areas of math produce homotopy with meaningful homotopy groups.

3.2 Long Exact Sequence of Homotopy Groups

Definition 3.7. A pointed Kan complex ia a Kan complex X together with a map $x : \Delta^0 \to X$. If $(X, x), (Y, y)$ are two pointed Kan complexes, we can define

$$A = \underline{\mathrm{Hom}}((X, x), (Y, y))$$

the Kan complex of pointed maps. An object in A is a pointed map. A morphism $\Delta^1 \to A$ a pointed homotopy. We can define pointed homotopy equivalence.

For each $n \geq 0$, here is a functor which takes every pointed Kan complexes $(X, x) \to \pi_n(X, x)$.

Remark. *$\pi_0(X, x)$ is a pointed set; $\pi_1(X, x)$ is a group; $\pi_n(X, x)$ is an abelian group if $n \geq 2$.*

Definition 3.8. Suppose $f(X, x) \to (Y, y)$ is a Kan fibration between pointed Kan complex. Then the fibre of f is the pullback $F = X \times_Y \Delta^0$.

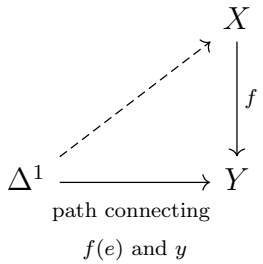
$$\begin{array}{ccc} F & \xrightarrow[i]{} & X \\ \downarrow & & \downarrow{\scriptstyle f} \\ \Delta^0 & \xrightarrow{y} & Y \end{array}$$

This is a pointed Kan complex.

Lemma 3.9. *There is an exact sequence of pointed sets*

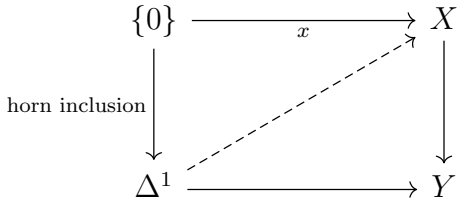
$$\pi_1(Y, y) \to \pi_0 F \xrightarrow{\pi_0 i} \pi_0 X \xrightarrow{\pi_0 f} \pi_0 Y$$

Proof. First we check the exactness at $\pi_0 X$. By construction we know $f \circ i$ is the constant map at y. It remains to show that for each $e \in X$ such that $f(e)$ is in the same path component of y, then e is connected to something in the fibre. To do this consider the following diagram:



It is clear that the lifted map is a path connecting e and some point in the fibre.

Next we check the exactne˙s at $\pi_0 F$. First of all we need to define the map $\pi_1(Y, y) \to \pi_0 F$:



Given a loop $\gamma : \Delta^1 \to Y$, the above diagram allows us to lift to a loop $\Delta^1 \to X$, which has end point in the fibre F_0. We define this to be the image of the loop $[\gamma] \in \pi_1(Y, y)$. The exactness is also clear, suppose $x' \in F_0$ is in the same connected component as x in X, then we can find a path connecting them in X, which maps to a path connecting them in Y, and the endpoints are both mapped to y. □

Remark. *If fact, there is a longer exact sequence:*

$$\cdots \to \pi_2(X, x) \to \pi_2(Y, y) \to \pi_1(F) \to \pi_1(X, x) \to \pi(Y, y) \to \pi_0 F \to \pi_0 X \to \pi_0 Y$$

3.3 Whitehead Theorem

Theorem 3.10 (Whitehead)**.** *Let $f : X \to Y$ be a map of Kan complexes. Then f is a homotopy equivalence if and only if:*

1. *$\pi_0 f = \pi_0 X \to \pi_0 Y$ is a bijection*
2. *For all $n \geq 1$, and all $x \in X_0$, $\pi_n f : \pi_n(X, x) \to \pi_n(Y, f(x))$ is an isomorphism of groups.*

Theorem 3.11 (Whitehead, pointed)**.** *A map $f(X, x) \to (Y, y)$ of* connected *pointed Kan complexes is an equivalcne of pointed Kan complexes iff $\pi_n(X, x) \to \pi_n(Y, y)$ is an iso for all $n \geq 1$.*

We now embark on the proof of Whitehead theorem.

Lemma 3.12. *If $f : X \to Y$ is a homotopy equivalence, then for all $x \in X_0$, $f : (X, x) \to (Y, f(x) = y)$ is a pointed homotopy equivalence. This proves the only if.*

Proof. Suffices to show for every pointed Kan complex (Z, z),

$$\pi_0\underline{\mathrm{Hom}}((Y, y), (Z, z)) \to \pi_0\underline{\mathrm{Hom}}((X, x), (Z, z))$$

is surjective. Assuming this, plug in $(Z, z) = (X, x)$, we get a map $g : (Y, y) \to (X, x)$ such that $g \circ f \simeq 1_{(X,x)}$. Since g is an unpointed homotopy equivalence, there exists h such that

$$h \circ g \simeq 1_{(Y,y)} \implies h \circ g \circ f \simeq f \implies h \simeq f$$

So g is a pointed homotopy inverse to f.

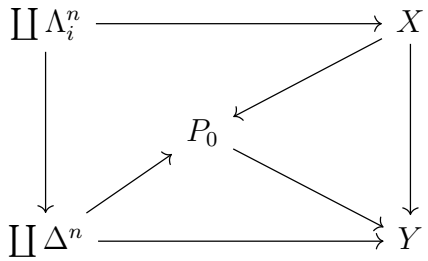
Now we prove the claim. Consider $\mathrm{ev}_x : \underline{\mathrm{Hom}}(X, Z) \to Z$, the fibre of this map at $z : \Delta^0 \to Z$ is $\underline{\mathrm{Hom}}((X, x), (Z, z))$. Hence we have a sequence:

$$\begin{array}{ccccccc}
\pi_1(Z,z) & \longrightarrow & \pi_0\underline{\mathrm{Hom}}((X,x),(Z,z)) & \longrightarrow & \pi_0\underline{\mathrm{Hom}}(X,Z) & \longrightarrow & \pi_0 Z \\
\uparrow{\scriptstyle =} & & \uparrow{\scriptstyle \text{surjective?}} & & \uparrow{\scriptstyle \simeq} & & \uparrow{\scriptstyle =} \\
\pi_1(Z,z) & \longrightarrow & \pi_0\underline{\mathrm{Hom}}((Y,y),(Z,z) & \longrightarrow & \pi_0\underline{\mathrm{Hom}}(Y,Z) & \longrightarrow & \pi_0 Z
\end{array}$$

The result follows from four lemma. □

Lemma 3.13 (Quillen's small object argument)**.** *Every map of simplicial sets is the composition of an anodyne map composed with a Kan fibration.*

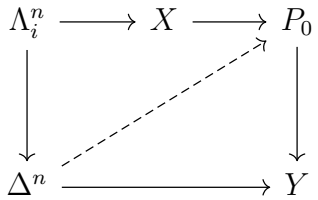
Proof. Suppose $X \to Y$ is a map of simplicial sets.



where P_0 is the push out. We have the factorization

$$X \to P_0 \to Y$$

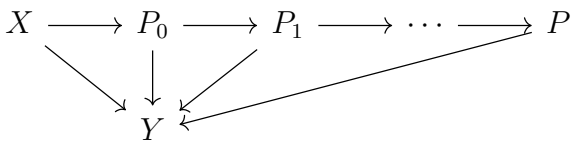
The first map is anodyne by construciton. Given a diagram



admits a lift. Apply the same procedure to $P_0 \to Y$ to get

$$X \to P_0 \to P_1 \to Y$$

Repeat this we get a diagram:



The map $X \to P$ as a transfinite composition of anodyne maps is anodyne. We claim that $P \to Y$ is a Kan fibration. □

Lemma 3.14. *As a set,* $\pi_n(X,x) \simeq \pi_0 \underline{\mathrm{Hom}}((\Delta^n, \partial\Delta^n),(X,x))$

Proof. Example for $n = 2$. There is a sequence of maps of simplicial sets

$$\partial\square^2 \to C \to \square^2$$

the first map is anodyne since its a pushout of a horn inclusion. Note $\square^2/C \simeq \Delta^2/\partial\Delta^2$ as pointed simplicial sets. Then

$$\underline{\mathrm{Hom}}((\square^2, \partial\square^2),(X,x)) \simeq \underline{\mathrm{Hom}}((\square^2/\partial\square^2, \{*\}),(X,x))$$

and

$$\underline{\mathrm{Hom}}((\Delta^2, \partial\Delta^2),(X,x)) \simeq \underline{\mathrm{Hom}}((\Delta^2/\partial\Delta^2, \{*\}),(X,x)) \simeq \underline{\mathrm{Hom}}((\square^2/C, \{*\}),(X,x))$$

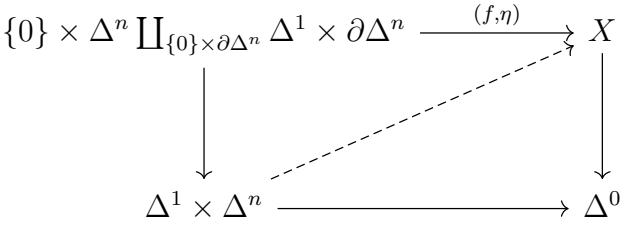
So it suffices to see $\underline{\mathrm{Hom}}((\square^2/C, \{*\}),(X,x)) \to \underline{\mathrm{Hom}}((\square^2/\partial\square^2, \{*\}),(X,x))$ is an equivalence. □

Lemma 3.15. *Suppose X is a connected Kan complex such that $\pi_n(X,x) \simeq 0$. Let $A \subset B$ be an inclusion of simplicial sets, $f : B \to X$ is a map and $\eta : \Delta^1 \times A \to X$ a homotopy from $f|_A$ to constant x. Then η can be extended to a homotopy:*

$$\overline{\eta} : \Delta^1 \times B \to X$$

from f to constant x.

Proof. It suffices to treat the case $A = \partial\Delta^n$ and $B = \Delta^n$ because the map $A \to B$ can be factored along cell attachment.



□

Lemma 3.16. *Suppose X is a connected Kan complex, and for each $n \geq 1$ and $x \in X$, $\pi_n(X, x) = 0$. Then the map $X \to \Delta^0$ is a trivial fibration.*

Proof. Suppose X is a connected Kan complex with trivial homotopy groups. It suffices to provide a lift:

$$\begin{array}{ccc} \partial\Delta^n & \xrightarrow{\sigma} & X \\ \downarrow & \nearrow & \\ \Delta^n & & \end{array}$$

Now apply the previous lemma with $A = \emptyset, B = \partial\Delta^n$, we get

$$\eta : \Delta^1 \times \partial\Delta^n \to X$$

from σ to constant x. Now conside the diagram:

$$\begin{array}{ccc} \Delta^1 \times \partial\Delta^n \coprod_{\{1\}\times\partial\Delta^n} \{1\} \times \Delta^n & \xrightarrow{(\eta, \mathrm{const}(x))} & X \\ \downarrow \text{anodyne} & \nearrow & \\ \{0\} \times \Delta^n & \longrightarrow & \Delta^1 \times \Delta^n \end{array}$$

giving a lift. □

We can now conclude the proof of Whitehead's theorem by the following lemma:

Lemma 3.17. *If $X \to Y$ is a Kan fibration between Kan complexes, satisfying the condtions in Whitehead's theorem, then it is a trivial fibration.*

Proof. Now suppose $f : X \to Y$ is a Kan fibration between Kan complexes, that is π_* isomorphic for all $x \in X_0$. We can use the long exact sequence to show the $\pi_* F \simeq 0$ for F the fibre. □

4 ∞-category of Spaces

This is also called "Anima".

4.1 First Construction

Let Kan be the fully subcategory of sSet with objects the Kan complexes. This is an ordinary category, but we can take nerve to upgrade Kan into an ∞-category. Now conisder

$$\coprod_{\text{homotopy equivalences of Kan complexes}} \Delta^1 \to \mathsf{Kan}$$

We push out against

$$\coprod_{\text{homotopy equivalences of Kan complexes}} \Delta^1 \to \coprod_{\text{homotopy equivalences of Kan complexes}} J$$

and get $\overline{\mathsf{Kan}} \in \mathsf{sSet}$. Use the small object argument to write

$$\overline{\mathsf{Kan}} \to \Delta^0$$

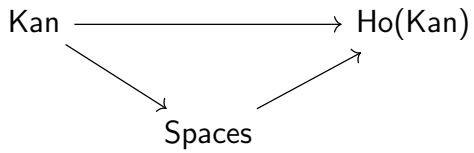
as the composite of an inner anodyne map and a map that lifts for inner horns

$$\overline{\mathsf{Kan}} \xrightarrow{\text{inner anodyne}} \mathsf{Spaces} \to \Delta^0$$

This gives Spaces as an ∞-category.

Proposition 4.1. *Suppose $\mathcal{C}$ is an ∞-category. Then the category $\underline{\mathrm{Hom}}(\mathsf{Spaces}, \mathcal{C})$ is equivalent to the full subcategory of $\underline{\mathrm{Hom}}(\mathsf{Kan}, \mathcal{C})$ that send all homotopy equivalences to isomorphisms.*

Corollary 4.2. *We have a factorization of ∞-categories:*



4.2 Second Construction

We can describe Spaces as an ∞-category by giving explicit simplices.

- The zero simplices are Kan complexes
- The one simplices are morphisms of Kan complexes
- The two simplices are: $\{X_0, X_1, X_2\}$ three Kan complexes, together with

  $$f : X_0 \to X_1; \quad g : X_1 \to X_2; \quad h : X_0 \to X_2$$

  and

  $$\Delta^1 \to \underline{\mathrm{Hom}}(X_0, X_2)$$

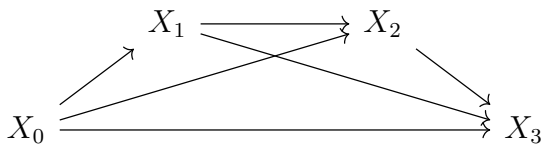
  a homotopy from $g \circ f$ to h.

  Remark. *The two simplices are $\{X_0, X_1, X_2\}$ three Kan complexes, together with*

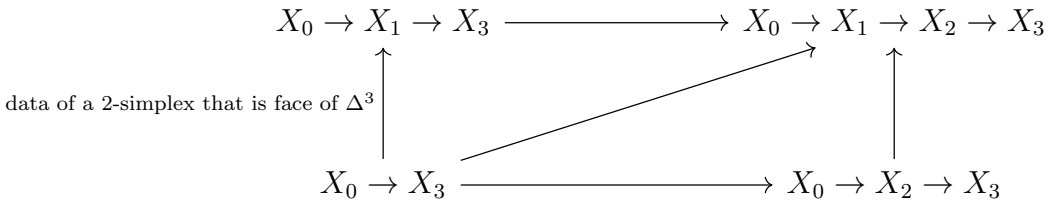
  $$f : X_0 \to X_1; \quad g : X_1 \to X_2; \quad h : X_0 \to X_2$$

  such that $g \circ f = h$.

- The three simplices are: $\{X_0, X_1, X_2, X_3\}$, maps between them, homotopics for every non-degenerate 2-simplex in Δ^3



, and finally a map $\Delta^1 \times \Delta^1 \to \underline{\mathrm{Hom}}(X_0, X_3)$ satisfying some conditions:



"homotopies between the homotopies that are the faces of Δ^3".

4.3 Third Construction

Definition 4.3. A *diagram* of spaces is a map $\mathcal{D} \to \mathsf{Spaces}$ for $\mathcal{D}$ an ∞-category. Given a diagram in spaces, we can form its colimit.

Proposition 4.4. *Suppose X is a Kan complex. There is a functor F:*

$$X \to \Delta^0 \xrightarrow{\text{picking out the unique object in } \Delta^0} \mathsf{Spaces}$$

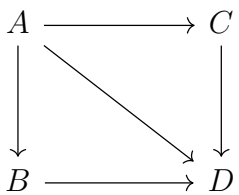
Then $X = \mathrm{colim}(F)$.

Definition 4.5. The ∞-category Spaces is characterized by the following property. Let $\mathcal{C}$ be a quasicategory with all colimits. A colimit preserving functor $\mathsf{Spaces} \to \mathcal{C}$ is the same data as an object of $\mathcal{C}$

Remark. *Compare in classical category theory, colimit preserving functor $\mathsf{Set} \to \mathcal{C}$ is the data of an object in $\mathcal{C}$.*

4.4 Homotopy Pullback

A *square* in Spaces, i.e. a map $\Delta^1 \times \Delta^1 \to \mathsf{Spaces}$ is a diagram



i.e. a square of Kan complexes that need not commute in Kan, together with a homotopy with the two composite.

Definition 4.6. A square

$$\begin{array}{ccc} P & \longrightarrow & C \\ \downarrow & & \downarrow \\ A & \longrightarrow & B \end{array}$$

is a *homotopy pullback square* if for any Kan complexe X

$$\pi_0 \underline{\mathrm{Hom}}(X, P)$$

is naturally isomorphic to

$$\pi_0 \underline{\mathrm{Hom}}((\Delta^1 \times \Delta^1, \Delta^0 \coprod \Lambda^2_2), (\mathsf{Spaces}, \begin{array}{ccc} X & & C \\ & & \downarrow \\ A & \longrightarrow & B \end{array}))$$

Lemma 4.7. *The homotopy pullback of a diagram:*

$$\begin{array}{ccc} & & B \\ & & \downarrow \\ A & \longrightarrow & C \end{array}$$

of Kan complexes is the Kan complex that is the pullback in Kan

$$\begin{array}{ccc} & & \underline{\mathrm{Hom}}(\Delta^1, C) \\ & & \downarrow \\ A \times B & \xrightarrow{(f,g)} & C \times C \end{array}$$

A point in the homotopy pullback is the data of a point in A, a point in B, and a path between their images in C.

Remark. *If A, B, C are pointed, so is their homotopy pullback.*

Example. *Suppose X is a pointed Kan complex, then ΩX is the homotopy pullback of*

$$\begin{array}{ccc} & & \Delta^0 \\ & & \downarrow \\ \Delta^0 & \longrightarrow & X \end{array}$$

Remark. $\Omega X \simeq \underline{\mathrm{Hom}}((\Delta^1, \partial\Delta^1), (X, \{x\}))$. *Thus* $\pi_0 \Omega X \simeq \pi_1 X$. *In fact,*

$$\pi_n \Omega X \simeq \pi_{n+1} X$$

for all n.

Example. *Let G be a group. There is a corresponding one-object groupoid. Call this BG. We can upgrade BG into a Kan complex by taking nerve. Then $\Omega BG = G$ is a dsjoint union of Δ^0. Thus:*

$$\pi_* BG = \begin{cases} 0, & \text{if } * \neq 1; \\ G, & \text{if } * = 1. \end{cases}$$

Remark. *Let G be the cyclic group of order 2, then $BG \simeq \mathbb{RP}^\infty$.*

Definition 4.8. If $f : A \to B$ is a map of pointed Kan complexes. The *homotopy fibre* of f is the homotopy pullback of

$$\begin{array}{ccc} & & A \\ & & \downarrow{\scriptstyle f} \\ \Delta^0 & \longrightarrow & B \end{array}$$

Remark. *There is a map from the homotopy fibre of f to A. Call this map g. Then: the homotopy fibre of g is just ΩB.*

$$\begin{array}{ccc} \Omega B = \mathrm{hfibre}(g) & \longrightarrow & \Delta^0 \\ \downarrow & & \downarrow \\ \mathrm{hfibre}(f) & \xrightarrow[g]{} & A \\ \downarrow & & \downarrow{\scriptstyle f} \\ \Delta^0 & \longrightarrow & B \end{array}$$

If $A \to B$ is a map between pointed Kan complexes, with homotopy fibre F, we can form

$$\cdots \to \Omega^2 B \to \Omega F \to \Omega A \to \Omega B \to F \to A \to B$$

Taking π_0, we get a sequence

$$\cdots \to \pi_2 B \to \pi_1 F \to \pi_1 A \to \pi_1 B \to \pi_0 F \to \pi_0 A \to \pi_0 B$$

Theorem 4.9. *This is an exact sequence of pointed sets.*

Theorem 4.10. *Suppose $f : A \to B$ is a Kan fibration between pointed Kan complexes, then the inclusion from the fibre of f to the homotopy fibre of f is an equivalence. In other words, the functor $F : \mathsf{Kan} \to \mathsf{Spaces}$ preserve the limits of the form*

$$\begin{array}{ccc} & & A \\ & & \downarrow{\scriptstyle \text{Kan fibration}} \\ C & \longrightarrow & B \end{array}$$

Remark. *A special case is that F preserves products.*

Remark. *The functor* $\mathsf{Kan} \to \mathsf{Spaces}$ *preserves pushouts of the form:*

$$\begin{array}{ccc} A & \xrightarrow{\text{mono}} & B \\ \downarrow & & \\ C & & \end{array}$$

In particular it preserves coproducts. Any $X \in \mathsf{Spaces}_0$ *is* $\mathrm{colim}_X \Delta^0$.

5 Simplicial R-modules

Definition 5.1. If R is a commutative ring, a simplicial R-module is a functor

$$\Delta^{\mathrm{op}} \to R\text{-mod}$$

Note a simplicial R-module X has the structure of a simplicial abelian group. Thus it is a Kan complex. It has a base point $0 \in X_0$.

Note if X_0 is a simplicial R-module, we can form a chain complex $C_*(X_\bullet)$, where $C_n(X_\bullet) = X_n$ and the boundary map: $\partial : X_n \to X_{n-1}$

$$\sigma \mapsto \sum_{i=0}^{n} (-1)^i d_i \sigma$$

One can check $\partial^2 = 0$.

Definition 5.2. If X is a simplicial set, define $\mathrm{Free}(X)$ be the simplicial R-module which on level n is the free R-module with basis X_n. Define $C_*(X; R)$ to be the chain complex associated to $\mathrm{Free}(X)$.

Remark. *If* X *is a topological space,* $C_*(\mathrm{Sing}(X); R)$ *is precisely the classical singular chain complex.*

Example. $C_*(\Delta^0; R)$ *is* R *in every degree:*

$$\cdots \to R \xrightarrow{1} R \xrightarrow{1} R \xrightarrow{0} R \to 0 \to 0 \to \cdots$$

Definition 5.3. If $A_\bullet$ is a simplicial R-module, we define $D_*(A_\bullet) \subset C_*(A_\bullet)$ to be the chain complex which in position n consists of R-linear combinations of degenerate n-simplices.

Proposition 5.4. *The boundary map* $\partial : C_n(A_\bullet) \to C_{n-1}(A_\bullet)$ *sends degenerate simplices to degenerate simplices.*

Definition 5.5. If $A_\bullet$ is a simplicial R-modules, then

$$N_*(A_\bullet) = C_*(A_\bullet)/D_*(A_\bullet)$$

This is called the *normalized chain complex*. On level n, it consists of nondegenerate n-simplices. For any simplicial set X, we can define $N_*(X; R)$.

Remark. *If X is a pointed Kan complex, then $N_*(X;R)$ has $N_*(\Delta^0;R)$ as a summand letting us define the "reduced normalized chain complex" $\widetilde{N}_*(X;R)$.*

Theorem 5.6. *If $X_\bullet$ is a simplicial R-module, then the projection $C_*(X_\bullet) \to N_*(X_\bullet)$ is a chain homotopy equivalences.*

Theorem 5.7 (Dold-Kan correspondence)**.** *The functor*

$$N_* : \{\text{simplicial } R\text{-modules}\} \to \mathrm{Ch}(R\text{-mod})_{\geq 0}$$

is an equivalence of categories.

Remark. *We can also form simplicial rings, simplicial groups, etc.*

Remark. *Suppose A is a simplicial abelian group. Then*

$$\pi_n A = \pi_0 \underline{\mathrm{Hom}}(\Delta^n/\partial\Delta^n, A) = \pi_0 \underline{\mathrm{Hom}}(\mathrm{Free}(\Delta^n/\partial\Delta^n), A) = \pi_0 \underline{\mathrm{Hom}}(\widetilde{N}_*(\Delta^n/\partial\Delta^n, \mathbb{Z}), \widetilde{N}(A))$$

Use Dold-Kan, we know this is

$$\pi_0 \underline{\mathrm{Hom}}(\cdots \to 0 \to \mathbb{Z} \to 0 \cdots, \widetilde{N}(A))$$

For the first chain concentrated in degree n. This is precisely $H_n(\widetilde{N}(A))$.

In other words, given a chain complex with know homology, we get a Kan complex with known homotopy groups.

Remark. *The inverse of N_* in the Dold-Kan correspondence is traditionally denoted as K.*

Definition 5.8. Suppose A is an abelian group, $n \geq 0$ an integere. Then we define

$$K(A,n) = K(\cdots \to 0 \to A \to 0 \to \cdots)$$

with A in degree n. This is called the *Eilenberg Maclane Space*.

Remark.

$$\pi_* K(A,n) = \begin{cases} A, & \text{if } * = n; \\ 0, & \text{otherwise.} \end{cases}$$

Example. *$K(A,0)$ has n-simplices*

$$\begin{aligned} \underline{\mathrm{Hom}}_{\mathsf{sSet}}(\Delta^n, K(A,0)) &= \underline{\mathrm{Hom}}_{\mathrm{sAb}}(\mathrm{Free}(\Delta^n), K(A,0)) \\ &= \mathrm{Hom}(N_*(\Delta^n), \cdots \to 0 \to A \to 0 \to \cdots) \\ &= A \end{aligned}$$

So $K(A,0) = \coprod_A \Delta^0$.

Example. *$K(A,1)$ has n-simplices consisting of elements $g_{ij} \in A$ for $0 \leq i \leq j \leq n$ satisfying conditions $g_{ij} + g_{jk} = g_{ik}$ for $0 \leq i < j < k \leq n$. This is BA.*

Definition 5.9. If X is a pointed set, define $K(X,0) = \coprod_X \Delta^0$. If G is a group, define $K(G,1) = BG$. If G is an abelian group and $n \geq 2$, we already defined $K(G,n)$.

Example. $K(C_2,1) \simeq \mathrm{Sing}(\mathbb{RP}^\infty)$; $K(\mathbb{Z},2) \simeq \mathrm{Sing}(\mathbb{CP}^\infty)$

Remark. *Suppose X is a Kan complex.*

$$\begin{aligned}\underline{\mathrm{Hom}}_{\mathsf{sSet}}(X, K(A,n)) &= \underline{\mathrm{Hom}}_{\mathrm{sAb}}(\mathrm{Free}(X), K(A,n)) \\ &= \underline{\mathrm{Hom}}_{\mathrm{ch}(\mathbb{Z})}(N_*(X), \cdots \to 0 \to A \to 0 \to \cdots)\end{aligned}$$

where A is at degree n. The set of maps i s the set of n-cocylces in $N^*(X;A)$.

$$\pi_0 \underline{\mathrm{Hom}}_{\mathrm{ch}(\mathbb{Z})}(N_*(X), \cdots \to 0 \to A \to 0 \to \cdots) = H^n(X;A)$$

In summary, for every abelian group A and integer $n \geq 0$, there is a Kan complex $K(A,n)$ such that

$$\pi_* K(A,n) \simeq \begin{cases} A, & \text{if } * = n; \\ 0, & \text{otherwise.} \end{cases}$$

$$\pi_0 \underline{\mathrm{Hom}}(X, K(A,n)) = H^n(X;A)$$

Note by Yoneda lemma, we see the second property characterizes $K(A,n)$ up to homotopy equivalence.

Remark. *There is a diagram of ∞-categories:*

$$\begin{array}{ccc} \mathsf{Kan} & \xrightarrow{\text{Free}} & \mathsf{sAb} \simeq \mathsf{Ch}(\mathbb{Z})_{\geq 0} \\ \downarrow & & \downarrow \\ \mathsf{Spaces} & \longrightarrow & D(\mathbb{Z})_{\geq 0} \end{array}$$

The bottom arrow is the unique colimit preserving functor sending Δ^0 to the chain of a single $\mathbb{Z}$ concentrated at level 0.

Remark. *Homology is best thought as the colimit preserving functor at the bottom.*

6 Spectral Sequence

6.1 Homological Spectral Sequence

Remark. *Suppose I have $0 \to C_* \to D_* \to D_*/C_*$, then there is an induced long exact sequence on cohomology. Now suppose I'm given a filtered chain complex:*

$$\begin{array}{ccccccccccc} 0 & \hookrightarrow & (C_1)_* & \hookrightarrow & (C_2)_* & \hookrightarrow & (C_3)_* & \hookrightarrow & \cdots & \to & C_* \\ & & \downarrow & & \downarrow & & \downarrow & & & & \\ & & (C_1)_* & & (C_2/C_1)_* & \xleftarrow[d_1]{} & (C_3/C_2)_* & & & & \end{array}$$

There are many long exact sequences. These are bonded in a spectral sequence. The question is how to calculate $H_*(C_*)$ in terms of $H_*(C_i/C_{i-1})$. We may view objects in $H_*(C_i/C_{i-1})$ are classes that might lift to elements of $H_*(C_i)$ that do not come from $H_*(C_{i-1})$.

Definition 6.1. A homological spectral sequence is a sequence of bigraded abelian groups:

$$E^1_{p,q}, E^2_{p,q}, E^3_{p,q}, \cdots$$

equipped with maps $d^r : E^r_{p,q} \to E^r_{p-r,q+r-1}$, and E^{r+1} is the homology of d^r. The E^∞-page is the colimit of E^r pages.

Remark. *In the above example, E^1-page consists of $H_*(C_i/C_{i-1})$.*

Theorem 6.2 (Serre Spectral Sequence)**.** *Let $F \to E \to B$ be a homotopy fibre sequence of pointed Kan complexes. Suppose B is simply connected ($\pi_0 B = \pi_1 B = 0$). Then there is a spectral sequence with*

$$E^2_{p,q} \simeq H_p(B, H_q(F, R))$$

converging to $H_{p+q}(E, R)$ in the sense that there is a filtration of $H_{p+q}(E, R)$ with associated graded isomorphic to $E^\infty_{p,q}$.

Remark. *Let $F \to E \to B$ be a homotopy fibre sequence. Assume B is connected, then changing the base point of B does not change the fibre up to equivalence.*

Remark. *Let $F \to F \times B = E \to B$ be a trivial homotopy fibre sequence. Then:*

$$H_n(E, R) = \bigoplus_{p+q=n} H_p(B, H_q(F, R)) \simeq \bigoplus H_p(B, R) \otimes_R H_q(F, R)$$

in good cases (for example if R is a field, or more generally $H_*(F, R)$ is a free R-module).

Example. *What is $H_*(\Omega S^3, \mathbb{Z})$? By definition, there is a homotopy fibre sequence*

$$\Omega S^3 \to \Delta^0 \to S^3$$

Since S^3 is simply connected, we can apply Serre spectral sequence

$$E^2 = H_p(S^3, H_q(\Omega S^3, \mathbb{Z})) \Rightarrow H_{p+q}(\Delta^0, \mathbb{Z})$$

The E^2-page is the same as the E^3-page, because d_2 differential lives in trivial groups. The E^4-page is the same as the E^∞-page.

q							
5							
4	$H_4 \simeq \mathbb{Z}$						
3	$H_3 \simeq 0$			$H_3 \simeq 0$			
2	$H_2(\Omega S^3) \simeq \mathbb{Z}$			$H_2(\Omega S^3) \simeq \mathbb{Z}$	0	0	
1	$H_1(\Omega S^3) \simeq 0$	0	0	$H_1(\Omega S^3) \simeq 0$	0	0	
0	$H_0(\Omega S^3) \simeq \mathbb{Z}$	0	0	$H_0(\Omega S^3) \simeq \mathbb{Z}$	0	0	
	0	1	2	3	4	5	p

$$H_0(\Omega S^3, \mathbb{Z}) \simeq \mathbb{Z}$$

$p + q = 0$ *contributs* $H_0(\Delta^0, \mathbb{Z})$

$$H_1(\Omega S^3, \mathbb{Z}) \simeq 0$$

becasue it is part of a filtration on $H_1(\Delta^0, \mathbb{Z}) = 0$.

$$H_2(\Omega S^3, \mathbb{Z}) \simeq \mathbb{Z}$$

d_3 *differential from* $\mathbb{Z}$ *to* $H_2(\Omega S^3)$ *must have no kernel and must be surjective in order for homology with respect to* d_3 *to leave the trivial group.*

Example. *What is* $H_*(K(\mathbb{Z}, 2), \mathbb{Z})$. *There is a homotopy fibre sequence*

$$S^1 = K(\mathbb{Z}, 1) \to \Delta^0 \to K(\mathbb{Z}, 2)$$

There is a spectral sequence

$$E^2_{p,q} = H_p(K(\mathbb{Z}, 2), H_q(S^1, \mathbb{Z})) \Rightarrow H_{p+q}(\Delta^0, \mathbb{Z})$$

We write $H_i = H_i(K(\mathbb{Z}, 2), \mathbb{Z})$.

q						
3	0	0	0	0	0	
2	0	0	0	0	0	
1	$H_0 = \mathbb{Z}$	$H_1 = 0$	$H_2 = \mathbb{Z}$	$H_3 = 0$	$H_4 = \mathbb{Z}$	
0	$H_0 = \mathbb{Z}$	$H_1 = 0$	$H_3 = \mathbb{Z}$	$H_3 = 0$	$H_4 = \mathbb{Z}$	
	0	1	2	3	4	p

d_2 *is the only possibly no ntrivial differential. So*

$$H_i(K(\mathbb{Z}, 2), \mathbb{Z}) = \begin{cases} \mathbb{Z}, & \text{if } i \text{ even;} \\ 0, & \text{otherwise.} \end{cases}$$

6.2 Construction of Serre Spectral Sequence

Quick construction

We fit in the pullback diagram into the following filtratoin:

$$\begin{array}{ccccccccccc} F & \longrightarrow & F_{(0)} & \longrightarrow & F_{(1)} & \longrightarrow & F_{(2)} & \longrightarrow & \cdots & \longrightarrow & E \\ \downarrow & & & & & & & & & & \downarrow \\ \Delta^0 & \longrightarrow & \mathrm{sk}_0 B & \longrightarrow & \mathrm{sk}_1 B & \longrightarrow & \mathrm{sk}_2 B & \longrightarrow & \cdots & \longrightarrow & B \end{array}$$

This gives rise to a filtration on $C_*(E, R)$ by $C_*(F_{(i)}, R)$.

More Serious Construction

The construction of Serre spectral sequence is the combination of the following two facts.

1. (Straightening correspondence / Grothendieck correspondence) If $F \to E \to B$ is a homotopy fibre sequence, with B connected. Then there exists a functor $G : B \to \mathsf{Spaces}$ taking each point $b \in B$ to F, such that $E = \mathrm{colim}(G)$

2. Now we can apply $C_* : \mathsf{Spaces} \to D(\mathbb{Z})_{\geq 0}$, to learn:

$$C_*(E) = \mathrm{colim}(B \xrightarrow{G} \mathsf{Spaces} \xrightarrow{C_*} D(\mathbb{Z})_{\geq 0})$$

   So now we need to compute $H_*(C_*(\mathrm{colim}(G))) = H_*(\mathrm{colim}(G))$.

The question is now how to compute the homology of a colimit. Given a chain complex C_*, we can form a filtration of it

$$\cdots \to \tau_{\geq 2} C_* \to \tau_{\geq 1} C_* \to \tau_{\geq 0} C_* = C_*$$

Here $\tau_{\geq i} C_*$ is the chain complex which looks like C_* itself for $* > i$, 0 for $* < i$, i-cycle in C_i at i (this is a "truncation"). Note

$$H_*(\tau_{\geq i} C_*) = \begin{cases} H_*(C_*), & \text{if } * \geq i; \\ 0, & \text{if } * < i. \end{cases}$$

This filtration gives rise to a spectral sequence with no differentials:

$$E^1\text{-page} = \oplus H_i(C_*)$$

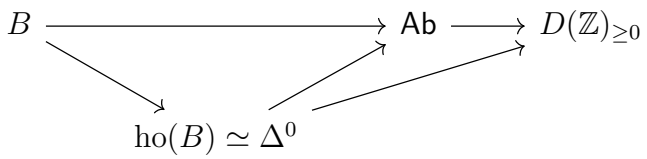
and all higher pages remain the same. Also note that $\tau_{\geq i} : D(\mathbb{Z})_{\geq 0} \to D(\mathbb{Z})_{\geq 0}$ is a functor.

Now $\operatorname{colim}(G)$ has a filtration $\operatorname{colim}(\tau_{\geq i} \circ G)$. The upshot is that we have a spectral sequence on E^1-page:

$$H_*(\operatorname{colim}(\tau_{\geq i} \circ G)/\operatorname{colim}(\tau_{\geq i+1} \circ G)) \Rightarrow H_*(\operatorname{colim}(G))$$

But we also know:

$$\operatorname{colim}(\tau_{\geq i} \circ G)/\operatorname{colim}(\tau_{\geq i+1} \circ G) = \operatorname{colim}(B \to \mathsf{Ab} \to D(\mathbb{Z})_{\geq 0})$$



where the first map is $b \mapsto H_i(G(b), \mathbb{Z})$, and the second map is stick the abelian group is position i and put 0 elsewhere.

Therefore, we have

$$\begin{aligned} H_*(\operatorname{colim}(\tau_{\geq i} \circ G)/\operatorname{colim}(\tau_{\geq i+1} \circ G)) &= H_*(\operatorname{colim}(B \to D(\mathbb{Z})_{\geq 0})) \\ &= H_*(B, H_i(G(b), \mathbb{Z})) \end{aligned}$$

6.3 Cohomological Spectral Sequence

Let $F \to E \to B$ for B simply connected. We have:

$$E_2^{p,q} = H^p(B; H^q(F; R))$$

and $E_\infty^{p,q}$ is the associated graded of a filtration on $H^{p+q}(E; R)$, and the differentials are pointing right and down:

$$d_r : E_r^{p,q} \to E_r^{p+r, q-(r-1)}$$

Suppowe we are in the situation such that

$$H^p(B; H^q(F; R)) \simeq H^p(B; R) \otimes_R H^q(F; R)$$

Then there is a cup product structure on the E_2 page:

$$(x \otimes y) \cup (x' \otimes y') = (-1)^{\deg y \cdot \deg x'}(x \cup x') \otimes (y \cup y')$$

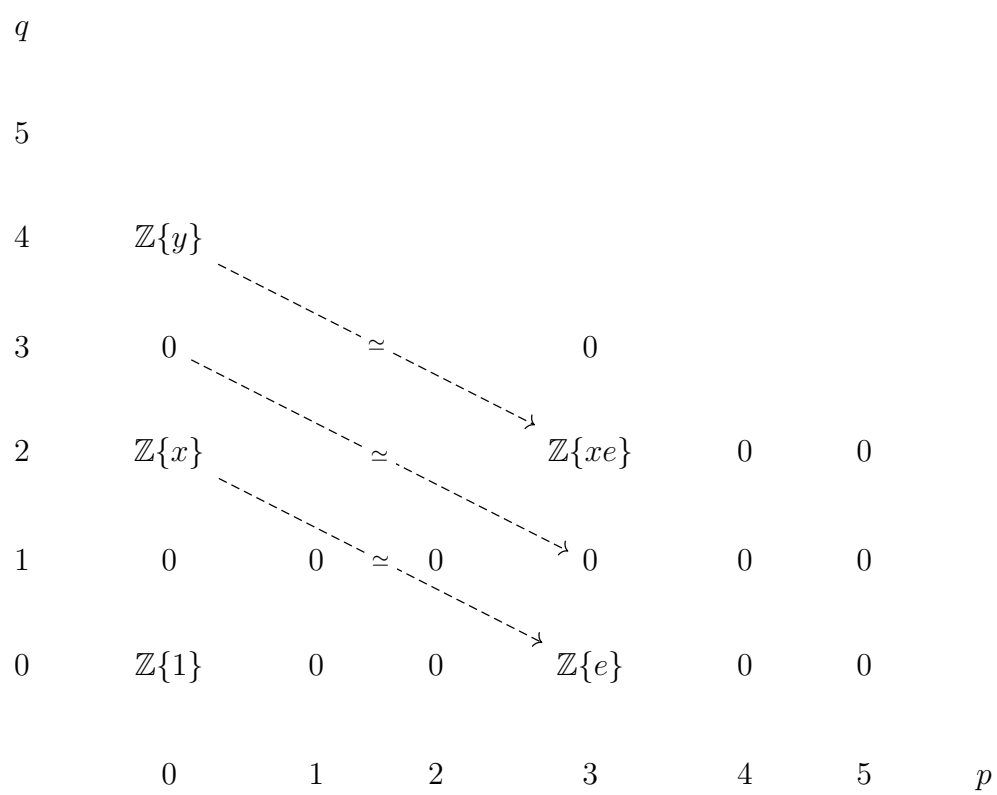
This induces a cup product structure in all higher pages. The cup product interacts with differentials in the following way: if $x \in E_r^{p,q}$, then

$$d_r(x \cup y) = d_r(x) \cup y + (-1)^{p+q} x \cup d_r(y)$$

Example. *We proved*

$$H_*(\Omega S^3; \mathbb{Z}) \simeq \begin{cases} \mathbb{Z}, & \text{if } * \text{ even;} \\ 0, & \text{otherwise.} \end{cases} \implies H^*(\Omega S^3; \mathbb{Z}) \simeq \begin{cases} \mathbb{Z}, & \text{if } * \text{ even;} \\ 0, & \text{otherwise.} \end{cases}$$

The Serre spectral sequence in cohomology is as follows:



Recall $H^*(S^3, \mathbb{Z}) = \mathbb{Z}[e]/e^2$. *We let* $x \in H^2(\Omega S^3; \mathbb{Z}), y \in H^4(\Omega S^3; \mathbb{Z}), z \in H^6(\Omega S^3; \mathbb{Z})$, *so*

$$d_3(x) = e, d_3(y) = xe, d_3(z) = ye$$

Note:

$$d_3(x^2) = d_3(x) \cdot x + x \cdot d_3(x) = 2xe \implies x^2 = 2y$$

$$d_3(xy) = d_3(x)y + xd_3(y) = ey + x(xe) = ye + 2ye = 3ye \implies xy = 3z$$
$$x^3 = x(2y) = 6z$$

In fact, we can compute:

$$H^*(\Omega S^3) = \mathbb{Z}[u_1, u_2, \cdots]/(u_i u_j = \binom{j+i}{i} u_{i+j})$$

Example. *Consider the fibre sequence* $\mathbb{CP}^\infty = K(\mathbb{Z}, 2) \to \Delta^0 \to K(\mathbb{Z}, 3)$.

$$E_2^{p,q} = H^p(K(\mathbb{Z}, 3); H^q(K(\mathbb{Z}, 2), \mathbb{Z})) \simeq H^p(K(\mathbb{Z}, 3); \mathbb{Z}) \otimes H^q(K(\mathbb{Z}, 2); \mathbb{Z})$$

q										
6	$\mathbb{Z}\{x^3\}$	0	0	$\mathbb{Z}\{x^3e\}$	0	0	$\mathbb{Z}/2\mathbb{Z}\{fx^3\}$	0	$\mathbb{Z}/3\mathbb{Z}\{gx^3\}$	
5	0	0	$\cdot 3$	0	0	0	0	0	0	
4	$\mathbb{Z}\{x^2\}$	0	0	$\mathbb{Z}\{x^2e\}$	0	0	$\mathbb{Z}/2\mathbb{Z}\{fx^2\}$	0	$\mathbb{Z}/3\mathbb{Z}\{gx^2\}$	
3	0	0	$\cdot 2$	0	0	0	0	0	0	
2	$\mathbb{Z}\{x\}$	0	0	$\mathbb{Z}\{xe\}$	0	0	$\mathbb{Z}/2\mathbb{Z}\{fx\}$	0	$\mathbb{Z}/3\mathbb{Z}\{gx\}$	
1	0	0	d_3	0	0	d_3	0	0	0	
0	$\mathbb{Z}\{1\}$	0	0	$\mathbb{Z}\{e\}$	0	0	$\mathbb{Z}/2\mathbb{Z}\{f\}$	0	$\mathbb{Z}/3\mathbb{Z}\{g\}$	
	0	1	2	3	4	5	6	7	8	p

We see

$$H^6(K(\mathbb{Z}, 3); \mathbb{Z}) \simeq \mathbb{Z}/2\mathbb{Z}$$

There is some non-zero element in this group, i.e. there is a map that is not null homotopoic

$$K(\mathbb{Z}, 3) \xrightarrow{f} K(\mathbb{Z}, 6)$$

So if X is any space, and $\alpha \in H^3(X; \mathbb{Z})$, we get a class in $H^6(X; \mathbb{Z})$:

$$X \xrightarrow{\alpha} K(\mathbb{Z}, 3) \xrightarrow{f} K(\mathbb{Z}, 6)$$

Indeed if $\alpha \in H^3(X; \mathbb{Z})$, $\alpha^2 = -\alpha^2 \implies \alpha^2$ is a 2-torsion.

Similarly, given any space X and any class in $H^3(X; \mathbb{Z})$, there is a cnanoical way to produce a 3-torsion class in $H^8(X; \mathbb{Z})$ (this is in fact the "Steenrod operations"), but there is no non-zero transformation $H^3(X; \mathbb{Z}) \to H^7(X; \mathbb{Z})$.

Assume we havre a homotpy fibre sequence $F \to E \to B$ for B simply connected. Recall on the E^2 page:

$$E^2_{p,q} = H_p(B; H_q(F; R)) \Rightarrow H_{p+q}(E; R)$$

On the E^∞-page, a quotient of $H_q(F; R)$ surrvives: this is the image of $H_q(F; R)$ in $H_q(E; R)$. Similarly, a sub of $H_p(B; R)$ survives: this is the image of $H_p(E; R)$ in $H_p(B; R)$.

6.4 Computation of Homotopy Group

Let X be a pointed Kan complex. We can compute π_0 and π_1.

Suppose X is simply connected. How to compute $\pi_2(X)$? It is the same as computing as $\pi_1(\Omega X) \simeq H_1(\Omega X; \mathbb{Z})$ (note ΩX is connected because X is simply connected). We can compute $H_1(\Omega X; \mathbb{Z})$ using

$$\Omega X \to \Delta^0 \to X$$

$$H_p(X; H_q(\Omega X; \mathbb{Z})) \Rightarrow H_{p+q}(\Delta^0; \mathbb{Z})$$

q						
3	$H_3(\Omega X)$					
2	$H_2(\Omega X)$					
1	$H_1(\Omega X)$					
0	$H_0(X)$	$H_1(X)$	$H_2(X)$	$H_3(X)$	$H_4(X)$	
	0	1	2	3	4	p

$$\Downarrow$$

q						
3	$H_3(\Omega X)$					
2	$H_2(\Omega X)$					
1	$H_1(\Omega X)$					
0	$\mathbb{Z}$	0 (iso)	$H_2(X)$	$H_3(X)$	$H_4(X)$	
	0	1	2	3	4	p

Hence if X is simply connected, $\pi_2(X) \simeq H_2(X; \mathbb{Z})$.

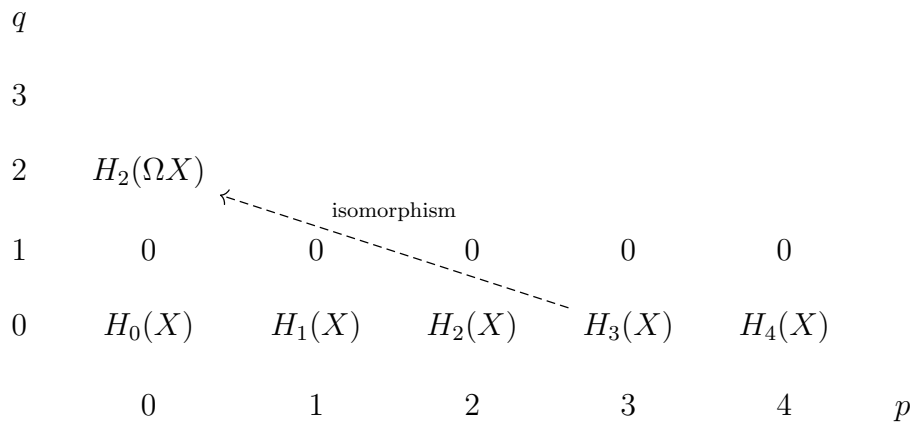
Theorem 6.3 (Hurewicz)**.** *Suppose X is a pointed Kan complex such that $\pi_* X = 0$ for $* < n$ and $n \geq 2$. Then:*

$$\pi_n X = H_n(X; \mathbb{Z})$$

In otherwords, the first nontrivial homotopy group agrees with the first nontrivial homology groups up to abelianization.

Proof. By induction on n. We do the example case for $n = 3$, assuming we know the statement for $n = 2$. Given X such that $\pi_0 X = \pi_1 X = \pi_2 X = 0$. Note that $\pi_3 X = \pi_2(\Omega X) = H_2(\Omega X; \mathbb{Z})$. It remains to show $H_2(\Omega X; \mathbb{Z}) \simeq H_3(X; \mathbb{Z})$. Again consider the fibre sequence $\Omega X \to \Delta^0 \to X$. We understand:

$$H_p(X; H_q(\Omega X)) \Rightarrow H_{p+q}(\Delta^0)$$



□

Question: What is $\pi_* S^3$?

We know $\pi_0(S^3) = \pi_1(S^3) = \pi_2(S^3) = 0$, and $\pi_3(S^3) = \mathbb{Z}$, by Hurewicz. We want to compute π_4.

Considerthe map $S^3 \to K(\mathbb{Z}, 3)$ that classifies a generator of $H^3(S^3; \mathbb{Z}) \simeq \mathbb{Z}$. On H^* this is an isomorphism for $* \leq 3$, this implies H_* this is an isomorphism for $* \leq 3$. By Hurewicz, this is an isomrphism for $* \leq 3$. We can take the homotopy fibre sequence:

$$F \to S^3 \to K(\mathbb{Z}, 3)$$

From the long exact sequence of homotopy groups:

$$\pi_* F \simeq \begin{cases} 0, & \text{if } * = 0, 1, 2, 3; \\ \pi_*(S^3), & \text{otherwise.} \end{cases}$$

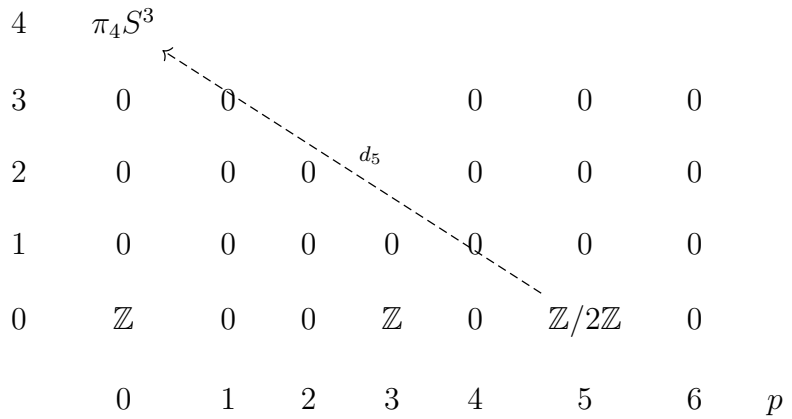
But then $\pi_4 F = H_4(F; \mathbb{Z})$.

Remark. *Last time we computed $H^*(K(\mathbb{Z}, 3); \mathbb{Z})$ in the range $* \leq 8$. This allows us to compute $H_*(K(\mathbb{Z}, 3); \mathbb{Z})$. We know:*

$$H_*(K(\mathbb{Z}, 3); \mathbb{Z}) = \begin{cases} \mathbb{Z}, & \textit{if } * = 0; \\ 0, & \textit{if } * = 1, 2; \\ \mathbb{Z}, & \textit{if } * = 3; \\ 0, & \textit{if } * = 4; \\ \mathbb{Z}/2\mathbb{Z}, & \textit{if } * = 5; \\ 0, & \textit{if } * = 6; \\ \mathbb{Z}/3\mathbb{Z}, & \textit{if } * = 7. \end{cases}$$

Now Let's look at the spectral sequence

$$H_p(K(\mathbb{Z},3);H_q(F)) \Rightarrow H_{p+q}(S^3;\mathbb{Z})$$



So $\pi_4 S^3 = \mathbb{Z}/2$.

6.5 Whitehead Tower

Suppose X is a pointed Kan complex. Let $\tau_{\geq 1}X$ denote the path component of the base point. We have $\tau_{\geq 1}X \to X$. This is a π_*-iso for $* > 0$, but $\pi_0(\tau_{\geq 1}X) = 0$

Let $\tau_{\geq 2}X$ denote the universal cover of $\tau_{\geq 1}$, and then $\tau_{\geq 2}X \to \tau_{\geq 1}X$ is a π_*-iso for $* > 1$, but $\pi_1(\tau_{\geq 2}X) \simeq 0$.

$\tau_{\geq 2}X$ is simply connected. Let $A = \pi_2(\tau_{\geq 2}X) \simeq H_2(\tau_{\geq 2}X;\mathbb{Z}) \simeq H^2(\tau_{\geq 2}X;\mathbb{Z})$. Hence we can provide a map

$$\tau_{\geq 2}X \to K(A,2)$$

that is an iso on π_2. We then let $\tau_{\geq 3}X$ denote the homotopy fibre of that map, and $\tau_{\geq 3}X \to \tau_{\geq 2}X$ is a π_*-iso for $* \geq 3$, but $\pi_2(\tau_{\geq 3}X) = 0$

Definition 6.4 (Whitehead Tower)**.** Associated to any pointed Kan complex X, there is a tower:

$$\begin{array}{ccc}
\tau_{\geq 4}X & \longrightarrow & K(\pi_4 X, 4) \\
\downarrow & & \\
\tau_{\geq 3}X & \longrightarrow & K(\pi_3 X, 3) \\
\downarrow & & \\
\tau_{\geq 2}X & \longrightarrow & K(\pi_2 X, 2) \\
\downarrow & & \\
\tau_{\geq 1}X & & \\
\downarrow & & \\
X & &
\end{array}$$

such that $\tau_{\geq n}X \to \tau_{\geq n-1}X \to K(\pi_{n-1}X, n-1)$ is a homotopy fibre sequence for each n. Moreover, we have

$$\pi_*(\tau_{\geq n}X) = \begin{cases} 0, & \text{if } * < n; \\ \pi_*X, & \text{otherwise.} \end{cases}$$

By Hurewicz, computing $H_n(\tau_{\geq n}X; \mathbb{Z})$ computes $\pi_n X$

Remark. *(for culture) Postnikov towers*

$$X \to \cdots \to \tau_{\leq 2}X \to \tau_{\leq 1}X \to \tau_{\leq 0}X$$

exists for any unpointed Kan complex X. If X is pointed, then $\tau_{\geq n+1}X = \text{hofiber}(X \to \tau_{\leq n}X)$. In fact, $\tau_{\leq n}X$ is the homotopy n-category of the Kan complex X.

Theorem 6.5 (Homology Whitehead Theorem)**.** *Suppsoe $f : X \to Y$ is a map of simply connected pointed Kan complexes. Then f is a homotopy equivalence iff $H_*(f; \mathbb{Z})$ is an isomorphism of graded groups.*

Proof. Let $F \to X \to Y$ be the homotopy fibre. To prove f is a homotopy equivalence, it suffices to prove $\pi_* f$ is an isomorphism, but this is equivalent to say $\pi_* F = 0$ for all $*$. But then by Hurewicz, this is equivalent to $H_* F \simeq 0$ for all $* > 0$ and F is simply connected.

But F is simply connected from the long exact sequence of homotopy groups. Therefore it suffices to show the vanishing of $H_* F$. We look at the Spectral sequence:

$$H_p(Y; H_q(F)) \Rightarrow H_{p+q}(X)$$

Recall on the E^2 page, the row $q = 0$ gives $H_p(Y)$. However, on the E^∞ page, what is left over is im $(H_p X \to H_p Y)$, which is everything. So there are no differentials off of this. all above parts of the E^2 page are actually 0. □

Corollary 6.6. *Suppose X is a simply connected Kan complex, such that*

$$\pi_i X \simeq \begin{cases} 0, & \text{if } i \neq 0; \\ A, & \text{if } i = n. \end{cases}$$

Then $X \simeq K(A, n)$. Therefore, Eilenberg-Maclane spaces are uniquely characterized (up to connected choice) by their homotopy groups.

Proof.

$$H_i(X, \mathbb{Z}) \simeq \begin{cases} 0, & \text{if } i < n; \\ A, & \text{if } i = n. \end{cases} \implies H^n(X; \mathbb{Z}) \simeq A$$

Then we constructed a map $X \to K(A, n)$ inducing π_* isomorphism. □

If X is any Kan complex, then $\pi_i X \simeq \pi_i \widetilde{X}$ for $\widetilde{X}$ universal cover for $i > 1$. We can compute $\pi_i \widetilde{X}$ by the Whitehead tower.

6.6 Mod $\mathcal{C}$ Theory

Let $\mathcal{C}$ be a subclass of abelian groups: either **finitely generated abelian groups**, or **finite abelian groups** that are $\mathcal{P}$**-torsion** for some subset $\mathcal{P}$ of prime numbers.

Remark. *Serre said "$\mathcal{C}$ is an ideal".*

Lemma 6.7. *Let $f : A \to B$ is a map of groups in $\mathcal{C}$. Then* $\ker(f)$ *and* coker (f) *are both in $\mathcal{C}$. Also, if $0 \to A \to C \to B \to 0$ and $A, B \in \mathcal{C}$, then $C \in \mathcal{C}$. Also, if $A, B \in \mathcal{C}$, then* $\mathrm{Tor}_i^{\mathbb{Z}}(A, B) \in \mathcal{C}$.

Lemma 6.8. *Suppose $F \to E \to B$ is a homotopy fibre sequence of pointed Kan complexes, such that B is simply connected. If any two of $\widetilde{H}_i(B;\mathbb{Z}), \widetilde{H}_i(E;\mathbb{Z}), \widetilde{H}_i(F;\mathbb{Z})$ are in $\mathcal{C}$ for all i, then so is the third.*

Remark. *A map $f : A \to B$ is said to be isomorphism mod $\mathcal{C}$ if the kernel and cokernal are both in $\mathcal{C}$.*

Proof. Assume $\widetilde{H}_i(B;\mathbb{Z}), \widetilde{H}_i(F;\mathbb{Z})$ are in $\mathcal{C}$ for all i. Because $\mathcal{C}$ is closed under extensions, to prove that $\widetilde{H}_{p+q}(E;\mathbb{Z})$ is in $\mathcal{C}$ for $p+q>0$, it suffices to prove $E^\infty_{p,q}$ is in $\mathcal{C}$ for $p, q > 0$. We know $\mathcal{C}$ is closed under formation of kernels and cokernels, so it suffices to prove $E^2_{p,q}$ is in $\mathcal{C}$ for $p, q > 0$. But then the short exact sequence

$$0 \to H_p(B) \otimes H_q(F) \to E^2_{p,q} = H_p(B; H_q(F)) \to \mathrm{Tor}(H_{p-1}(B), H_q(F)) \to 0$$

gives the desired result. □

Lemma 6.9. *If $A \in \mathcal{C}$, then $H_i(K(A,n);\mathbb{Z})$ is in $\mathcal{C}$ for all $i, n > 0$.*

Proof. If $n \geq 2$, we have

$$K(A, n-1) \to \Delta^0 K(A, n)$$

with $K(A,n)$ simply connected. So it suffices to prove the result for $n = 1$. We need to understand $H_i(K(A,1);\mathbb{Z})$ for $i > 0$ (this is the group homology of group A with constant module $\mathbb{Z}$).

Note if A is finitely generated, then $A = \mathbb{Z}^{\oplus r} \oplus (\mathbb{Z}/2^{a_1}) \oplus \cdots (\mathbb{Z}/2^{a^n}) \oplus (\mathbb{Z}/3^{b_1}) \oplus \cdots$, so

$$K(A,1) = K(\mathbb{Z},1)^{\times r} \times K(\mathbb{Z}/2^{a_1}, 1) \times \cdots$$

Note $K(\mathbb{Z},1) = S^1$. Hence it suffices to prove:
Claim: $H_i(K(\mathbb{Z}/p^n, 1);\mathbb{Z})$ is finite and p-power torsion.
Proof of Claim: Consider $K(\mathbb{Z},1) = S^1 \to F \to K(\mathbb{Z},2) \xrightarrow{p^n} K(\mathbb{Z},2)$ inducing multiplication by p^n on π_2. From the long exact sequence we know

$$\pi_i F = \begin{cases} 0, & \text{if } i \neq 1; \\ \mathbb{Z}/p^n, & \text{if } i = 1. \end{cases}$$

So $F = K(\mathbb{Z}/p^n, 1)$, and we get

$$S^1 \to K(\mathbb{Z}/p^n, 1) \to K(\mathbb{Z}, 2)$$

we can use cohomological spectral sequence to compute the cohomology of $K(\mathbb{Z}/p^n, 1)$, and then switch back to homology. □

Theorem 6.10 (Mod $\mathcal{C}$ Hurewicz)**.** *Suppose X is simply connected, and $\pi_i(X) \in \mathcal{C}$ for $0 < i < n$, then*

$$\pi_n X \to H_n(X; \mathbb{Z})$$

is an isomorphism mod $\mathcal{C}$.

Proof. Consider the Whitehead tower

$$\begin{array}{ccc}
\tau_{\geq n}X & & \\
\downarrow & & \\
\cdots & & \\
\downarrow & & \\
\tau_{\geq 4}X & \longrightarrow & K(\pi_4 X, 4) \\
\downarrow & & \\
\tau_{\geq 3}X & \longrightarrow & K(\pi_3 X, 3) \\
\downarrow & & \\
\tau_{\geq 2}X = X & \longrightarrow & K(\pi_2 X, 2)
\end{array}$$

We know $\pi_n X = H_n(\tau_{\geq n}X; \mathbb{Z})$. We need to prove:

$$H_n(\tau_{\geq n}X; \mathbb{Z}) \to H_n(\tau_{\geq 2}X; \mathbb{Z}) = H_n(X; \mathbb{Z})$$

is an isomorphism mod $\mathcal{C}$.

Suppose $\tau_{\geq j}X \to \tau_{\geq j-1}X \to K((\pi_{j-1}X; j-1))$ where $\pi_{j-1}X \in \mathcal{C}$. Then:

$$H_p(K(\pi_{j-1}X, j-1); H_q(\tau_{\geq j}X; \mathbb{Z})) \Rightarrow H_{p+q}(\tau_{\geq j-1}X; \mathbb{Z})$$

and $H_p(K(\pi_{j-1}X, j-1); \mathbb{Z}) \in \mathcal{C}$. This will eventually implies $H_q(\tau_{\geq j}X; \mathbb{Z})$ is isomorphic to $H_{p+q}(\tau_{j-1}X; \mathbb{Z})$ mod $\mathcal{C}$ □

Corollary 6.11. *$\pi_i(S^n)$ is finitely generated for all i and $n \geq 2$*

Proof. Take $\mathcal{C}$ to be the class of finitely generated abelian groups and apply mod $\mathcal{C}$ Hurewicz. □

Corollary 6.12. *For $i > 3$, then $\pi_i(S^3)$ is finite.*

Proof. Consider $F \to S^3 \to K(\mathbb{Z}; 3)$. Note $\pi_i(S^3)$ is finite for all $i > 3$, then it suffices to check $\pi_i F$ is finite for all i. By Hurewicz, it suffices to check $H_i(F; \mathbb{Z})$ is finite for all i. But the homology of $H_i(F; \mathbb{Z})$ is finitely generated, we only need to show $H_i(F; \mathbb{Z}) \otimes \mathbb{Q} = 0$. By UCT, we only need to show $H_i(F; \mathbb{Q}) \simeq 0$. The rest is omitted. □

7 Homotopy Group of Spheres

7.1 Homotopy Pushout

There are three ways to think of Spaces:

1. Kan[h.e^{-1}]
2. Top[h.e.$^{-1}$]
3. sSet[w.e.$^{-1}$]

Given

$$\begin{array}{ccc} & & C \\ & & \downarrow \\ A & \longrightarrow & B \end{array}$$

in Kan, the homotopy pullback (pullback in Spaces) is the pullback in sSet

$$\begin{array}{ccc} & & B^{\Delta^1} = \mathrm{Hom}(\Delta^1, B) \\ & & \downarrow \\ A \times C & \longrightarrow & B \end{array}$$

For homotopy pushout: suppose

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \\ C & & \end{array}$$

The homotopy pushout (pushout in Spaces) is the pushout

$$\begin{array}{ccc} A \coprod A & \longrightarrow & B \coprod C \\ \downarrow & & \\ A \times \Delta^1 & & \end{array}$$

up to weak equivalence. Note the homotopy pushout of Kan complexes need not be a Kan complex.

Remark. *If*

$$\begin{array}{ccc} X & \xrightarrow{f} & Z \\ {\scriptstyle g}\downarrow & & \\ Y & & \end{array}$$

is a diagram in Top*, then*

$$X \times [0,1] \coprod Y \coprod Z / (x \times \{0\} \simeq f(x), x \times \{1\} \simeq g(x))$$

is the homotopy pushout, identified under the functor Top $\to$ Spaces *by inverting homotopy equivalences.*

Definition 7.1. Suppose X is a pointed topological space, pointed by $x \in X$. The suspension X

$$\Sigma X = X \times [0,1]/(X \times \{0,1\}, \{x\} \times [0,1])$$

is a pointed space.

There is a suspension isomorphism:

$$H_*(\Sigma X) \simeq H_{*-1}(X)$$

Remark. *ΣX is the homotopy pushout in Spaces_* (the ∞-category of pointed Spaces):*

$$\begin{array}{ccc} X & \longrightarrow & \Delta^0 \\ \downarrow & & \downarrow \\ \Delta^0 & \longrightarrow & \Sigma X \end{array}$$

Example. *Note that:*

$$\underline{\mathrm{Hom}}(\Sigma S^n, X) = \underline{\mathrm{Hom}}\left(\begin{array}{ccc} S^n & \longrightarrow & \Delta^0 \\ \downarrow & & \\ \Delta^0 & & \end{array}, X\right) = \begin{array}{ccc} & & \underline{\mathrm{Hom}}_{\mathsf{Spaces}}(\Delta^0, X) \\ & & \downarrow \\ \underline{\mathrm{Hom}}_{\mathsf{Spaces}}(\Delta^0, X) & \longrightarrow & \underline{\mathrm{Hom}}_{\mathsf{Spaces}}(S^n, X) \end{array}$$

But this is just the pullback:

$$\begin{array}{ccc} & & \Delta^0 \\ & & \downarrow \\ \Delta^0 & \longrightarrow & \Omega^n X \end{array} = \Omega^{n+1} X = \underline{\mathrm{Hom}}(S^{n+1}, X)$$

Hence by Yoneda, we have $\Sigma S^n \simeq S^{n+1}$.

Remark. *It is clear that we have an isomorphism in Spaces:*

$$\underline{\mathrm{Hom}}(\Sigma X, Y) \simeq \underline{\mathrm{Hom}}(X, \Omega Y)$$

for any $Y \in \mathsf{Spaces}_*$. In fact, we have: this isomorphism holds in the 1-category of sSet.

Note: there exists a natural map, for any $X \in \mathsf{Spaces}_*$, $X \to \Omega\Sigma X$. Thus there is a map $\pi_i X \to \pi_i \Omega\Sigma X = \pi_{i+1}(\Sigma X)$.

Theorem 7.2 (Freudenthal)**.** *Suppose X is a pointed space and $\pi_i X = 0$ for $i < n$. Then:*

$$\pi_i(X) \to \pi_{i+1}(\Sigma X)$$

is an isomorphism for $i \leq 2n-2$ and is surjection for $i = 2n-1$.

Remark. *We thus have:*

$$\pi_3 S^2 \simeq \pi_4 S^3 \simeq \pi_5 S^4 \simeq \cdots$$

We write $\pi_1^{\text{st}} = \mathbb{Z}/2\mathbb{Z}$ *to mean* $\pi_{n+1}S^n \simeq \mathbb{Z}/2\mathbb{Z}$ *for* $n \gg 0$. *Similarly, we define* $\pi_k^{\text{st}} = \pi_{n+k}S^n$ *for* $n \gg 0$.

Proof. Consider

$$\Omega\Sigma X \to \Delta^0 \to \Sigma X$$

$$E_2 = H_p(\Sigma X; H_q(\Omega\Sigma X; \mathbb{Z}))$$

But recall $H_p(\Sigma X; A) = H_{p-1}(X; A) = 0$ for $p \leq n$. Note also $H_q(\Omega\Sigma X; \mathbb{Z})$ vanishes for q small beacuse $\pi_q(\Omega\Sigma X) = \pi_{q+1}(\Sigma X)$ does.

The differential from one edge of the spectral sequence to the other is an isomorphisms:

$$H_{i-1}(x) = H_i(\Sigma X) \to H_{i-1}(\Omega\Sigma X)$$

for $i \leq\approx 2n$ With care, we see that $H_{i-1}(X) \to H_{i-1}(\Omega\Sigma X)$ iso for $i \leq\approx 2n$, the map is: $H_{i-1}(X \to \Omega\Sigma X)$.

Examine $F \to X \to \infty\Omega\Sigma X$, we see that $H_*(F; \mathbb{Z})$ is trivial for $* \leq\approx 2n \implies \pi_* F$ is trivial for $* \leq\approx 2n \implies \pi_* \to \pi_*\Omega X$ is an isomorphism for $* \leq\approx 2n$. □

7.2 Recent Progress on Stable Homotopy Groups

Here is a list of stable homotopy groups for spheres:

π_i^{st}	$\mathbb{Z}$	$\mathbb{Z}/2\mathbb{Z}$	$\mathbb{Z}/2\mathbb{Z}$	$\mathbb{Z}/24\mathbb{Z}$	0	0	$\mathbb{Z}/2\mathbb{Z}$	$\mathbb{Z}/240\mathbb{Z}$	$\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$
i	0	1	2	3	4	5	6	7	8

Remark. *Serre's method can be streamlined into a single spectral sequence for every prime* p, *called the* **Adams spectral sequence***:*

$$E_2 = \text{Ext}_{\mathcal{A}}(\mathbb{F}_p, \mathbb{F}_p) \Longrightarrow (\pi_*^{\text{st}})_{(p)}$$

where $\mathcal{A}$ *is a graded ring, the* **Steenrod algebra** *(this is the cohomology of Eilenberg-Maclane spaces* $K(\mathbb{Z}/p\mathbb{Z}, n)$ *in the "stable range".)*

The top of the E_∞*-page displays* im $(J) \subset \pi_*^{st}$

Theorem 7.3 (Lin-Wang-Xu, 2024)**.** h_6^2 *survives to the* E_∞*-page, and so represents a class in* $(\pi_{126}^{\text{st}})_{(2)}$.

There is a map $J : \text{O}(n) \to \Omega^n S^n = \underline{\text{Hom}}_{\mathsf{Spaces}_*}(S^n, S^n)$ via one point compactification. So we have a map:

$$\pi_k \text{O}(n) \to \pi_k \Omega^n S^n = \pi_{k+n} S^n = \pi_k^{\text{st}}$$

for $n \gg 0$. im (J) entirely understood by Quillen, Adams, Sullivan in early 70s. These groups are given by numerators of Bernoulli numbers.

Theorem 7.4 (Kervaire-Milnor)**.** *There is an exact sequence of abelian groups, for every $n \geq 5$,*

$$0 \to bP_{n+1} \to \Theta_n \to \pi_n^{\mathrm{st}}/\mathrm{im}\ (J) \to KI \to 0$$

where: Θ_n is the group of exotic n-spheres up to diffeomorphism (group structure is connected sum); bP_{n+1} is the group of exotic spheres that are boundaries of framed manifolds (this turns out to be a cyclic group of order known in terms of denominators of Bernoulli numbers); KI is either 0 or $\mathbb{Z}/2\mathbb{Z}$.

Theorem 7.5 (Browder)**.** *KI is the trivial group unless $n = 2^m - 2$ for some m. In dimension $2^{j+1} - 2$, it is $\mathbb{Z}/2\mathbb{Z}$ iff a class h_j^2 survives to the E_∞-page of the 2-primary Adams Spectral sequence.*

Remark. *KM proved it is $\mathbb{Z}/2\mathbb{Z}$ if $n = 2, 4, 14$. Lin-Wang-Xu proved it is $\mathbb{Z}/2\mathbb{Z}$ if $n = 126$. Mahowald-Tangora-Jones proved it is $\mathbb{Z}/2\mathbb{Z}$ if $n = 30, 62$.*

Theorem 7.6 (Hill-Hopkins-Raverel, 2016)**.** *KI is trivial in all dimensions > 126.*

Remark. *This uses chromatic homotpy at height 4.*

Question: On E_∞-page, how many total dots exist in the first n columns?

Partial Answer: The number of dots on E_2-page is approximately $n^{\log(n)^2}$. This is the best known upper bound; The number of dots on E_∞-page $\geq$ number of dots in im (J) approximately $\sqrt{n}$. Oka in the 80s prove the number of dots on E_∞-page has lower bound approximately n (produced β family element controlled by modular forms). In 2023, it was proved it has lower bound approximately $n\log(n)$.

8 Cohomology of Eilenberg-Maclane Spaces

We focus on $H^*(K(\mathbb{F}_2); \mathbb{F}_2)$. In particular, $H^i(K(\mathbb{F}_2, n); \mathbb{F}_2)$ is homotopy classes of maps

$$K(\mathbb{F}_2, n) \to K(\mathbb{F}_2, i)$$

thus represents a natural transformation

$$H^n(X; \mathbb{F}_2) \to H^i(X; \mathbb{F}_2)$$

Construction:
There are natural transformations $\mathrm{Sq}^i : H^*(X; \mathbb{F}_2) \to H^{*+i}(X; \mathbb{F}_2)$ for $i \geq 0$ satisfying:

1. $\mathrm{Sq}^0 = \mathrm{id}$
2. $\mathrm{Sq}^i(x) = x^2$ if $i = |x|$
3. $\mathrm{Sq}^i(x) = 0$ if $i > |x|$
4. Sq^i is a group homomorphism: $\mathrm{Sq}^i(a + b) = \mathrm{Sq}^i(a + b) = \mathrm{Sq}^i(a) + \mathrm{Sq}^i b$
5. $\mathrm{Sq}^k(ab) = \sum_{i+j=k} \mathrm{Sq}^i(a)\mathrm{Sq}^j(b)$

6. Sq^i commutes with $\Sigma : \tilde{H}^k(X;\mathbb{F}_2) \simeq \tilde{H}^{k+1}(\Sigma X;\mathbb{F}_2)$

For $n > 0$, $\mathrm{Sq}^n \iota_n = \iota_n^2 \in H^{2n}(K(\mathbb{F}_2,n);\mathbb{F}_2)$, where $\iota_n \in H^n(K(\mathbb{F}_2,n);\mathbb{F}_2)$ is the element corresponding to the identity map $K(\mathbb{F}_2,n) \to K(\mathbb{F}_2,n)$.

Loo at Serre spectral sequence $K(\mathbb{F}_2,n) \to \Delta^0 \to K(\mathbb{F}_2,n+1)$.

$$E_2 = H^p(K(\mathbb{F}_2,n+1);\mathbb{F}_2) \otimes_{\mathbb{F}_2} H^q(K(\mathbb{F}_2,n);\mathbb{F}_2)$$

when ι is still on the page, then $d(\iota^2) = 0$. This implies the image of $\mathrm{Sq}^n \iota_n = \iota_n^2 \in H^{2n}(K(\mathbb{F},2);\mathbb{F}_2)$ in $H^{2n+1}(K(\mathbb{F}_2,n+1);\mathbb{F}_2)$ survives, we define this image to be $\mathrm{Sq}^n \iota_{n+1} \in H^{2n+1}(K(\mathbb{F}_2,n+1);\mathbb{F}_2)$.

Example. $H^*(K(\mathbb{F}_2,1);\mathbb{F}_2) = H^*(\mathbb{RP}^\infty;\mathbb{F}_2) \simeq \mathbb{F}_2[x]$ *with* $|x| = 1$.

1. $\mathrm{Sq}^1(x) = x^2$
2. $\mathrm{Sq}^i(x) = 0$ *for* $i > 1$
3. $\mathrm{Sq}^1(x^2) = 0$
4. $\mathrm{Sq}^2(x^2) = x^4$
5. $\mathrm{Sq}^1(x^3) = x^4$
6. $\mathrm{Sq}^2(x^3) = x^5$
7. $\cdots$

In fact, $\mathrm{Sq}^i(x^k) = \binom{k}{i} x^{k+i}$

Suppose $F \to E \to B$ is a fibre sequence and B simply connected, and F connected.

$$E_2 = H^p(B;\mathbb{F}_2) \otimes_{\mathbb{F}_2} H^q(F;\mathbb{F}_2)$$

Classes on the bottom row that survive the spectral sequence give im $(H^*(B;\mathbb{F}_2) \to H^*(E;\mathbb{F}_2))$; classes on the first column survive the spectral sequence give im $(H^*(E;\mathbb{F}_2) \to H^*(F;\mathbb{F}_2))$

A pair $(x,y) \in H^{i+1}(B;\mathbb{F}_2) \times H^i(F;\mathbb{F}_2)$ is called **transgressive** if

$$\begin{array}{ccc} H^{i+1}(E,F) & \xleftarrow{\quad\delta\quad} & H^i(F) \\ \uparrow{\scriptstyle p} & & \\ \tilde{H}^{i+1}(B) = H^{i+1}(B,\{b\}) & & \end{array}$$

$p(x) = \delta(y)$.

Definition 8.1. If $F \to E$ is any map of Kan complexes, we define $H^{i+1}(E,F)$ to be H^{i+1} of the homotopy pushout of

$$\begin{array}{ccc} F & \longrightarrow & E \\ \downarrow & & \\ \Delta^0 & & \end{array}$$

Proposition 8.2. *A pair (x, y) is transgressive iff x, y surviving until the E_{i+1}-page and $d_{i+1}(y) = x$.*

Corollary 8.3 (Kudo Transgression)**.** *Suppose $F \to E \to B$, B simply connected. Then if $y \in H^{i+1}(F; \mathbb{F}_2)$ survives to E_{i+1} and $d_{i+1}(y) = x$, then $\mathrm{Sq}^n(y)$ survives to the E_{i+n+1}-page and $d_{i+n+1}(\mathrm{Sq}^n(y)) = \mathrm{Sq}^n(x)$.*

Consider $K(\mathbb{F}_2, 1) \to \Delta^0 \to K(\mathbb{F}_2, 2)$ and try to compute $H^*(K(\mathbb{F}, 2); \mathbb{F}_2)$. The spectral sequence is formal given the Leibniz rule and Kudo transgression theorem.

Theorem 8.4 (Serre)**.**

$$H^*(K(\mathbb{F}_2, 2); \mathbb{F}_2) = \mathbb{F}_2[\iota_2, \mathrm{Sq}^1\iota_2, \mathrm{Sq}^2\mathrm{Sq}^1\iota_2, \mathrm{Sq}^4\mathrm{Sq}^2\mathrm{Sq}^1\iota_2, \cdots]$$

Example. *$K(\mathbb{F}_2, 1) \times K(\mathbb{F}_2, 1) \to K(\mathbb{F}_2, 2)$ the cup product of two degree 1 classes. On cohomology we have:*

$$\mathbb{F}_2[\iota_2, \mathrm{Sq}^1\iota_2, \mathrm{Sq}^2\mathrm{Sq}^1\iota_2, \mathrm{Sq}^4\mathrm{Sq}^2\mathrm{Sq}^1\iota_2, \cdots] \to \mathbb{F}_2[x, y]$$

$$\iota_2 \mapsto xy, \quad \mathrm{Sq}^1\iota_2 \mapsto \mathrm{Sq}^1(xy) = x^2y + xy^2, \cdots$$

This map is injective on degree ≤ 4. As an application, we know $\mathrm{Sq}^1\mathrm{Sq}^1\iota_2 \mapsto \mathrm{Sq}^1\mathrm{Sq}^1(xy) = 0 \implies \mathrm{Sq}^1\mathrm{Sq}^1\iota_2 = 0$.

Theorem 8.5 (Serre)**.** *$H^*(K(\mathbb{F}_2, n); \mathbb{F}_2)$ is a polynomial ring with generators*

$$\mathrm{Sq}^{i_1}\mathrm{Sq}^{i_2} \cdots \mathrm{Sq}^{i_j}\iota_n$$

such that $i_k - 2i_{k+1} \geq 0$ for all k and $\sum(i_k - 2i_{k+1}) < n$,

Theorem 8.6. *For any space X, the following relations hold in $H^*(X; \mathbb{F}_2)$:*
(Adam relation) if $i < 2j$, $\mathrm{Sq}^i\mathrm{Sq}^j = \sum_{k=0}^{\frac{i}{2}} \binom{j-k-1}{i-2k} \mathrm{Sq}^{i+j-k}\mathrm{Sq}^k$

Example.

- $\mathrm{Sq}^1\mathrm{Sq}^1 = 0$
- $\mathrm{Sq}^3 = \mathrm{Sq}^1\mathrm{Sq}^2$
- $\mathrm{Sq}^5 = \mathrm{Sq}^1\mathrm{Sq}^4$
- $\mathrm{Sq}^6 = \mathrm{Sq}^2\mathrm{Sq}^4 + \mathrm{Sq}^5\mathrm{Sq}^1$

Remark. *The Steenrod algebra $\mathcal{A}$ is the associate ring with generators Sq^i and Adam relations. The Adams Spectral Sequence starts with $\mathrm{Ext}_{\mathcal{A}}(\mathbb{F}_2, \mathbb{F}_2)$*

Remark. *$\mathcal{A}$ also appears in the theory of formal groups.*

There is an analogue of all of this for $\mathbb{F}_p$.

9 Introduction to Spectra

There are quasi-categories:

$$\mathsf{Set} = \{\text{discrete Spaces}\} \hookrightarrow \mathsf{Spaces}$$

and

$$R\text{-mod} \hookrightarrow D(R)_{\geq 0}$$

Note Spaces is a category generated under colimits by Δ^0, we can thus define a free functor $\mathsf{Spaces} \to D(R)_{\geq 0}$. A further step is to define other "algebraic" quasi-categories.

9.1 $\mathbb{E}_1$-Spaces

Definition 9.1. A discrete monoid is a set X with multiplication $\mu : X \times X \to X$, a unit element $e \in X$ with the property that multiplication is unital and associative.

We wish to define the upgraded version of Monoid, namely $\mathbb{E}_1$-spaces. As a Heurestic, $\mathbb{E}_1$-spaces is a space X, a multiplication $X \times X \to X$, an identity element $e : \Delta^0 \to X$. For each $x, y, z \in X$, a 1-simplex from $(xy)z$ to $x(yz)$; and higher data.

Definition 9.2. Given an ∞-groupoid X, an $\mathbb{E}_1$-structure on X is a functor

$$M : \mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Spaces}$$

such that $M([0]) = \Delta^0, M([1]) = X$. The injective maps $[1]$ to $[n]$ of form $i \leq i+1$ produce a map

$$M([n]) \to M([1])^{\times n}$$

which is an equivalence.

Example. *Suppose X is a classical monoid. Then $M : \mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Set} \to \mathsf{Spaces}$ as the nerve of the one object category X.*

Example. *Consider any topological group G. I have $\mathrm{Sing}(G)$ a $\mathbb{E}_1$-space:*

$$\mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Top} \to \mathsf{Spaces}$$

Example. *If X is a Kan complex, then $\underline{\mathrm{Hom}}(X, X)$ has a natural $\mathbb{E}_1$-structure.*

Example. *If X is a Kan complex, then ΩX has a natural $\mathbb{E}_1$-structure. Composition of loops is associative and unital up to* **coherent** *homotopy.*

Note if X is a space, $\pi_0(\Omega X)$ is a group. In general if X is an $\mathbb{E}_1$-monoid, $\pi_0(X)$ is just a monoid.

Definition 9.3. An $\mathbb{E}_1$-space X is *group like* if $\pi_0 X$ is a group.

Remark. $\mathbb{E}_1$-spaces *is a full subcategory of* $\underline{\mathrm{Hom}}(\mathbf{\Delta}^{\mathrm{op}}, \mathsf{Spaces})$. *Inside of this, we have the fully faithful subcategory of group like $\mathbb{E}_1$-spaces. Being group like is a property of $\mathbb{E}_1$-spaces, not an extra data.*

Definition 9.4. If $M : \mathbf{\Delta}^{\mathrm{op}} \to \mathsf{Spaces}$ is an $\mathbb{E}_1$-space. Then BM is the colimit of M.

Example. *If G is a discrete group, then $BG = K(G, 1)$ is a one object groupoid. If M is a discrete monoid, $BM = BG$ for G the "group-completion" of M.*

Definition 9.5. A *vector bundle* on a space X is a map of simplicial sets: $X \to BU(n)$.

Theorem 9.6. *The functors:*

$$B : (\mathbb{E}_1\text{-}\mathsf{spaces})^{\text{gp like}} \to (\mathsf{Spaces}_*)_{\geq 1}$$

where $(\mathsf{Spaces}_*)_{\geq 1}$ is the full subcategory of Spaces_* on connected pointed space. Also I have:

$$\Omega : (\mathsf{Spaces}_*)_{\geq 1} \to (\mathbb{E}_1\text{-}\mathsf{spaces})^{\text{gp like}}$$

These are inverse equivalence of quasi-categories.

Corollary 9.7 (Seifert-van Kampen)**.**

$$\pi_1 : (\mathsf{Spaces}_*)_{\geq 1} \to \mathsf{Groups}$$

preserves pushouts

Proof. $\pi_1 X = \pi_0 \Omega X$. Under the above equivalence, this is

$$\pi_0 : (\mathbb{E}_1\text{-}\mathsf{spaces})^{\text{gp like}} \to \mathsf{Groups}$$

It remains to show π_0 preserves pushout. □

Corollary 9.8. *ΩBM is the group completion of M, whenever M is an $\mathbb{E}_1$-space.*

Proof. For any group like $\mathbb{E}_1$-space G, we have

$$\underline{\mathrm{Hom}}_{\mathbb{E}_1\text{-}\mathsf{spaces}}(M, G) \simeq \underline{\mathrm{Hom}}_{\mathbb{E}_1\text{-}\mathsf{spaces}}(M^{\text{gp completion}}, G) \simeq \underline{\mathrm{Hom}}(\Omega BM^{\text{gp completion}}, G)$$

So B is insensitive to group completion. □

Corollary 9.9. *The free group like $\mathbb{E}_1$-space on a pointed space X is $\Omega\Sigma X$*

Example. *If X is a group like $\mathbb{E}_1$-space, then $\pi_n X$ can be thought of $\mathbb{E}_1$-space map from $\Omega S^{n+1} = \Omega\Sigma S^n \to X$ up to equivalence.*

9.2 $\mathbb{E}_\infty$-Spaces

If $\mathcal{C}$ is any category with products, we can define $\mathbb{E}_1$-algebra in $\mathcal{C}$ as functors $F : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{C}$ such that $F([0]) = \prod_\emptyset$, and the maps $F([n]) \to F([1])^{\times n}$ are isomorphisms. So $\mathbb{E}_1$-spaces is $\mathbb{E}_1$-algbera with $\mathcal{C} = \mathsf{Spaces}$.

In particular, the category $\mathbb{E}_1$-spaces has products. We can thus define an $\mathbb{E}_2$-space is an $\mathbb{E}_1$-algebra in $\mathbb{E}_1$-space, an $\mathbb{E}_3$-space is an $\mathbb{E}_1$-algebra in $\mathbb{E}_2$-spaces, etc.

Note: an $\mathbb{E}_k$-space has k compatible $\mathbb{E}_1$-space structures. For example, $\mathbb{E}_2$-spaces ($\mathbb{E}_1$-algebras in $\mathbb{E}_1$-spaces) has two forgetufl functors to $\mathbb{E}_1$-spaces. EH argument says these two underlying $\mathbb{E}_1$ spaces of any $\mathbb{E}_2$ space are equivalent and π_0 of the $\mathbb{E}_2$ space is commutative.

Definition 9.10. An $\mathbb{E}_k$ space X is group like if $\pi_0 X$ is a group.

Theorem 9.11 (May-Recognition Theorem)**.** *The category of group-like $\mathbb{E}_k$ spaces is equivalent to the category of k-fold loop spaces $(\mathsf{Spaces}_*)_{\geq k}$. We have*

$$\Omega^k : (\mathsf{Spaces}_*)_{\geq k} \to \mathbb{E}_k^{\text{gp like}}\text{-spaces}$$

an equivalence of categories.

When $k = 1$, this says "groups are the data of a one-object groupoid".

Definition 9.12. An $\mathbb{E}_\infty$ space X is a space with a collection of compatible $\mathbb{E}_k$ structures for all $k \geq 1$. A group like $\mathbb{E}_\infty$ space is the data of a sequence of spaces $X_0, X_1, X_2, \cdots$ together with equivalences $X_{i-1} \simeq \Omega X_i$. We also require $X_i \in (\mathsf{Spaces}_*)_{\geq i}$.

Remark. *People say "∞-loop spaces" for group like $\mathbb{E}_\infty$-spaces.*

Example. *Say A is a discrete abelian group. Then*

$$A = \Omega K(A, 1) = \Omega^2 K(A, 2) = \Omega^3 K(A, 3) = \cdots$$

Hence abelian groups are discrete ∞-loops spaces. Any discrete abelian monoid is an $\mathbb{E}_\infty$ space.

Example. *$\Omega^2 S^3$ is $\mathbb{E}_2$ but not $\mathbb{E}_\infty$. Note $\Omega^2 S^3 = \Omega^3 BSU(2)$, so is in fact an $\mathbb{E}_3$ space.*

Example. *Any symmetric monoidal classical groupoid is an E_∞-space. For example, consider the category of finite sets with morphisms being bijections. This is:*

$$\coprod_{n \geq 0} B\Sigma_n$$

for Σ_n is the symmetric group of order $n!$. This is an E_∞-space under disjoint union of finite sets.

Example. *Suppose R is a classical commutative cring. Then the category of finite generated free R-modules, with isomorphisms as morphisms. This is an $\mathbb{E}_\infty$-space under $\oplus$.*

Example. *There is an $\mathbb{E}_\infty$ space $\coprod_{n \geq 0} BU(n)$. The groupoid of finite dimensional $\mathbb{C}$-vector space. The $\mathbb{E}_\infty$-structure comes from $\oplus$.*

Remark. *$M \mapsto \Omega BM$ is the right adjoint to the embedding of group like $\mathbb{E}_\infty$ spaces into $\mathbb{E}_\infty$ spaces.*

Example.

1. *If $M = \coprod BU(n)$, then $\Omega BM = BU \times \mathbb{Z}$. This is "virtual vector spaces"*
2. *If M is finitely generated free R-modules. Then $\pi_i \Omega BM$ are known as algebraic K-groups of R.*
3. *If M is finite sets with disjoint union. Then $\pi_k(\Omega BM) = \pi_k^{\text{st}}$.*

Example. *One can construct MO an infinite loop space.*

- *Objects are* 0*-dim compact manifolds*
- *Morphisms are cobordism of* 0*-dim manifolds.*
- 2*-morphisms are cobordisms of* 1*-morphisms*

This is $\mathbb{E}_\infty$ *under disjoint union of manifolds. The base point is the empty manifold.* $\pi_k MO$ *are closed* k*-manifolds up to cobordism.*

Example. *If* X *is a pointed space, there are maps*

$$X \to \Omega\Sigma X \to \Omega^2\Sigma^2 X \to \Omega^3\Sigma^3 X \to \cdots$$

The colimit $\Omega^\infty\Sigma^\infty X$ *is an* $\mathbb{E}_\infty$*-space,* $\pi_i\Omega^\infty\Sigma^\infty X = \pi_i^{\mathrm{st}} X$*.*

9.3 Spectra

This is another analogue of abelian groups.

Definition 9.13. A **spectrum** is a sequence $(X_0, X_1, \cdots)$ of pointed spaces and equivalences $X_i \simeq \Omega X_{i+1}$. Recall if each X_i is i-connected, then we say the spectrum is an ∞-loop spaces.

Example. *Recall we can upgrade every discrete abelian group into a spectrum.*

Example. *(Bott-Periodicity)* $\Omega(BU \times \mathbb{Z}) \simeq U$*,* $\Omega U = BU \times \mathbb{Z}$*. Hence we have a spectrum:*

$$KU = (BU \times \mathbb{Z}, U, BU \times \mathbb{Z}, U, \cdots)$$

Example.

$$\Delta^0 \to \Omega\Delta^0 \to \Omega^2\Delta^0 \to \cdots$$

This is denoted as 0*. This is both the initial and terminal spectrum.*

Example. *If* X *is a space,*

$$X_0 = \mathrm{colim}(X \to \Omega\Sigma X \to \Omega^2\Sigma^2 X \to \cdots)$$

is a group like $\mathbb{E}_\infty$*-space.*

Suppose $\mathcal{C}$ is a category with 0 object and finite limits. Then $\Omega : \mathcal{C} \to \mathcal{C}$ is the functor

$$c \mapsto \lim\left(\begin{array}{ccc} & & 0 \\ & & \downarrow \\ 0 & \longrightarrow & c \end{array}\right)$$

If $\mathcal{C}$ is the spectra, and

$$\Omega : (X_0, X_1, \cdots) \mapsto (\Omega X_0, X_0, \cdots)$$

this is an equivalence of category: I can recover all lower terms by taking loops.

Definition 9.14. An ∞-category $\mathcal{C}$ is *stable* if it has finite limit, a 0 object, and $\Omega_{\mathcal{C}}$ is an auto equivalence.

Example. *The category of Spectra is stable.*

Example. *Consider $D(\mathbb{Z})$ the category of chain complexes with quasi-isomprhisms inverted. This is stable.*

Proposition 9.15. *Suppose $\mathcal{C}$ is a stable category. Then $\mathcal{C}$ has finite colimits. If we define*

$$\Sigma_{\mathcal{C}} : c \mapsto \operatorname{colim}\left(\begin{array}{ccc} c & \longrightarrow & 0 \\ \downarrow & & \\ 0 & & \end{array} \right)$$

Then $\Sigma_{\mathcal{C}}$ is the inverse to $\Omega_{\mathcal{C}}$. Furthermore, a square (with filling)

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

is a pullback in $\mathcal{C}$ iff it is a pushout in $\mathcal{C}$.

This follows from the following lemma:

Lemma 9.16. *Let $\mathcal{C}$ be a stable quasicategory. The forgetful functor from the subcategory of pullback square $\underline{\mathrm{Hom}}(\Delta^1 \times \Delta^1, \mathcal{C}) \to \underline{\mathrm{Hom}}(\Lambda^2_0, \mathcal{C})$ is an equivalence of quasi-categories.*

Proof. Suppose we start from a pullback square:

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

The forgetful functor gives us:

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \\ C & & \end{array}$$

But then we can recover the pullback square via the following diagram:

$$\begin{array}{ccccccc} \Omega A & \longrightarrow & \Omega C & & & & \\ \downarrow & \lrcorner & \downarrow & & & & \\ \Omega B & \longrightarrow & X & \longrightarrow & F & \longrightarrow & 0 \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ 0 & \longrightarrow & G & \longrightarrow & A & \longrightarrow & B \\ & & \downarrow & \lrcorner & \downarrow & & \\ & & 0 & \longrightarrow & C & & \end{array}$$

□

Suppose $X = (X_0 \xrightarrow{\sim} \Omega X_1 \xrightarrow{\sim} \Omega^2 X_2 \xrightarrow{\sim} \cdots)$ is a spectrum. Then we define $\pi_i X \coloneqq \pi_{i+j} X_j$ whenever $i + j \geq 0$. This is always an abealin group. Here i can be a negative integer. If X is an ∞-loop space, then $\pi_i X = 0$ for $i < 0$.

Theorem 9.17 (Whitehead)**.** *A map $X \to Y$ of spectra is an equivalence iff $\pi_i X \to \pi_i Y$ is an isomorphism of abelian groups whenever $i \in \mathbb{Z}$.*

A sequence of objects $A \to B \to C$ in a stable category is said to be a *fibre sequence* or *cofibre sequence* if there eixts a square

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & C \end{array}$$

that is a pushout / pullback.

A fibre sequence of spectra gives rise to an exact sequence

$$\cdots \pi_i A \to \pi_i B \to \pi_i C \to \pi_{i-1} A \to \pi_{i-1} B \to \cdots$$

infinite in both directions. This is a consequence of the fact that

$$\Omega^\infty : \mathsf{Spectra} \to \mathsf{Spaces}_*, \quad (X_0, X_1, \cdots) \mapsto X_0$$

preserves limits. The left adjoint is $\Sigma^\infty : \mathsf{Spaces}_* \to \mathsf{Spectra}$, $X \mapsto \operatorname{colim}(X \to \Omega\Sigma X \to \cdots)$

Remark. *A cofibre sequence of pointed spaces is a push out*

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ \Delta^0 & \longrightarrow & C \end{array}$$

There is a LES for stable homotopy groups.

Notation: Given $X \in \mathsf{Spaces}$, we write $X_+ \in \mathsf{Spaces}_*$ as the image of the left adjoint of the forgetful functor from Spaces_* to Spaces. We also write $\Sigma^\infty_+ X \coloneqq \Sigma^\infty(X_+)$ for unpointed X.

Remark. *Σ^∞ takes values in the subcategory of spectra E such that $\pi_i E = 0$ for all $i < 0$. This subcategory is the category of group like $\mathbb{E}_\infty$ spaces.*

Example. *If X is a pointed space,*

$$\pi_i \Sigma^\infty X = \pi_i^{\mathrm{st}} X = \pi_i(\operatorname{colim}_n \Omega^n \Sigma^n X) = \operatorname{colim}_n \pi_i \Omega^n \Sigma^n X$$

Example. *If A is a discrete abelian group, it is a spectrum. This is called an* Eilenberg MacLane spectrum.

Recall that Spectra is stable (pushout squares $\iff$ pullback squares), has limits and colimits (not necessarily finite), has zero object $0 = \Sigma^\infty \Delta^0$, has Σ and $\Sigma^{-1} = \Omega$ as inverse autoequivalence.

Remark. *If A is a discrete abelian group, and n an integer, then $\Sigma^n A$ is also called an Eilenberg-Maclane spectrum*

Example. *$\Omega^\infty \Sigma^n A = K(A, n)$ for $n \geq 0$. It equals Δ^0 if $n < 0$*

Another feature is spectra has an internal $\underline{\mathrm{Hom}}$. If E, F are spectra,

$$\underline{\mathrm{Hom}}(E, F) = \Omega \underline{\mathrm{Hom}}(E, \Sigma F) = \cdots$$

Remark. *If $\mathcal{C}$ is any stable category, then tehre is a spectrum of maps betwen any two objects. Namely:*

$$\underline{\mathrm{Hom}}_{\mathcal{C}}(c_1, c_2) \in \text{Spectra}$$

9.4 Cohomology and Homology

Suppose $E = (E_0, E_1, \cdots)$ is a spectrum, and X is a pointed space.

Definition 9.18. The n-th E-cohomology group of X is defined as:

$$\widetilde{E}^n(X) = \pi_{-n} \underline{\mathrm{Hom}}_{\mathsf{Spectra}}(\Sigma^\infty X, E) = \pi_0 \underline{\mathrm{Hom}}_{\mathsf{Spectra}}(\Sigma^\infty X, \Sigma^n E) = \pi_0 \underline{\mathrm{Hom}}_{\mathsf{Spaces}_*}(X, \Omega^\infty \Sigma^n E)$$

Example. *If $E = (A, K(A,1), K(A,2), \cdots)$. Then*

$$\widetilde{E}^n(X) = \pi_0 \underline{\mathrm{Hom}}_{\mathsf{Spaces}_*}(X, K(A, n)) = \widetilde{H}^n(X, A)$$

Example. *If $E = KU$, then* $\widetilde{E}^i(X) = \begin{cases} [X, BU \times \mathbb{Z}], & \text{if } i \text{ even;} \\ [X, U], & \text{otherwise.} \end{cases}$

Remark. *Let E be a spectrum. Then $X \mapsto \underline{\mathrm{Hom}}_{\mathsf{Spectra}}(\Sigma^\infty X, E)$ sends cofibre sequences to cofibre sequences. Hence $\widetilde{E}^*(X)$ satisfies Mayer-Vietoris.*

Theorem 9.19 (Brown representability)**.** *Suppose given a sequence of functors*

$$\widetilde{E}^n : (\mathsf{Spaces}_*)_{\geq 1} \to \mathsf{Ab}$$

for $n \in \mathbb{Z}$ satisfying Mayer-Vietoris, there exists a spectrum E such that

$$\widetilde{E}^n = \pi_{-n} \underline{\mathrm{Hom}}(\Sigma^\infty X, E)$$

Definition 9.20. If X is an unpointed space, $E^* X \coloneqq \widetilde{E}^*(X_+)$

Remark. *If X is pointed, then $E^* X = \widetilde{E}^* X \oplus E^*(\Delta^0)$.*

Remark. *We have $\Sigma^\infty \Delta^0 = 0$, but $\Sigma^\infty_+ \Delta^0 = \Sigma^\infty(\Delta^0_+) = \Sigma^\infty S^0 = \mathbb{S}$ is the sphere spectrum. $\mathbb{S}$ for spectra is the analogue of $\mathbb{Z}$ for abelian groups. We define $S^n \coloneqq \Sigma^n \mathbb{S} = \Sigma^\infty S^n = \Sigma^\infty \Sigma^n S^0$. We can also also define S^n for n negative (this will not be Σ^∞ of pointed space).*

Remark. *One can computed* $\pi_0 \underline{\mathrm{Hom}}_{\mathsf{Spectra}}(S^n, E) = \pi_n E$

Remark. *If* X *is a spectrum, and* E *is another spectrum, I define*

$$\widetilde{E}^* X \coloneqq \pi_0 \underline{\mathrm{Hom}}_{\mathsf{Spectra}}(X, \Sigma^* E)$$

Question: What is the $\mathbb{F}_2$ cohomology of $\mathbb{F}_2$?

Lemma 9.21. *Suppose* E *is a group like* $\mathbb{E}_\infty$ *space, thought of as a spectrum. Then:*

$$E = \mathrm{colim}_n \Sigma^{-n} \Sigma^\infty \Omega^\infty \Sigma^n E$$

Proof. We have $\Omega^\infty \Sigma^n E$ has first homotopy group in degree n. By Freudenthal, we see stable homtopy groups of $\Omega^\infty \Sigma^n E$ agree with the homotopy group up to degree $2n-1$. □

So we can compute:

$$\begin{aligned}
\mathbb{F}_2^* \mathbb{F}_2 &= \mathbb{F}_2^*(\mathrm{colim}_n \Sigma^{-n} \Sigma^\infty \Omega^\infty \Sigma^n \mathbb{F}_2) \\
&= \lim_n \mathbb{F}_2^*(\Sigma^{-n} \Sigma^\infty \Omega^\infty K(\mathbb{F}_2, n)) \\
&= \lim_n \pi_0 \underline{\mathrm{Hom}}(\Sigma^{-n} \Sigma^\infty K(\mathbb{F}_2, n), \Sigma^* \mathbb{F}_2) \\
&= \lim_n \pi_0 \underline{\mathrm{Hom}}(\Sigma^\infty K(\mathbb{F}_2, n), \Sigma^{*+n} \mathbb{F}_2) \\
&= \lim_n H^{*+n}(K(\mathbb{F}_2, n), \mathbb{F}_2)
\end{aligned}$$

As a graded $\mathbb{F}_2$ vector space, this is the Steenrod algebra $\mathcal{A}$ with Sq^2 in $\mathbb{F}_2^2 \mathbb{F}_2$.

Remark. *If* A, B *are spectra, we can form* $A \otimes B$ *another spectrum. This is characterized as:*

$$\underline{\mathrm{Hom}}(A \otimes B, C) = \underline{\mathrm{Hom}}(A, \underline{\mathrm{Hom}}(B, C))$$

To show existence, by adjoint functor theorem it suffices to check $\underline{\mathrm{Hom}}(B, -)$ *preserves limits. Note:*

$$\underline{\mathrm{Hom}}(A \otimes \mathbb{S}, C) \simeq \underline{\mathrm{Hom}}(A, \underline{\mathrm{Hom}}(\mathbb{S}, C)) \simeq \underline{\mathrm{Hom}}(A, C)$$

Thus $\mathbb{S}$ *is the unit for* $\otimes$.

Remark. *If* X *and* Y *are unpointed spaces, then*

$$\underline{\mathrm{Hom}}(X \times Y, Z) = \underline{\mathrm{Hom}}(X, \underline{\mathrm{Hom}}(Y, Z))$$

Then $\Sigma^\infty_+ X \times Y \simeq \Sigma^\infty_+ X \otimes \Sigma^\infty_+ Y$. *If* E *and* F *are spectra with no negative* π_* *we can use the lemma to express* $E \otimes F$ *in terms of desuspensions of* $\otimes$ *of suspension spectrum.*

Remark. *If* A *and* B *are abelian groups, then* $A \otimes^{\mathbb{L}}_{\mathbb{Z}} B \neq A \otimes_{\mathbb{S}} B$. *The later can have many nonvanishing homotopy groups.*

Definition 9.22. If E is a spectrum, and X is a pointed space. We define the E-homology of X:

$$\widetilde{E}_*X = \pi_*(E \otimes_{\mathbb{S}} \Sigma^\infty X)$$

If X is a unpointed:

$$E_*X = \pi_*(E \otimes_{\mathbb{S}} \Sigma^\infty_+ X)$$

If X is a spectrum:

$$E_*X = \pi_*(E \otimes_{\mathbb{S}} X)$$

Stable homotopy is an example of homology, with $E = \mathbb{S}$.

Example. *Up to homotopy, we have maps*

$$\mathbb{F}_2 \to \Sigma^i \mathbb{F}_2$$

are given degree i elements of $\mathcal{A}$.

Example. *The fibre F of $\mathbb{F}_2 \xrightarrow{\mathrm{Sq}^1} \Sigma\mathbb{F}_2$ has*

$$\pi_* F = \begin{cases} \text{an extension of } \mathbb{F}_2 \text{ by } \mathbb{F}_2, & \text{if } * = 0; \\ 0, & \text{otherwise.} \end{cases}$$

This implies F is an Eilenberg Maclane spectrum, either $\mathbb{Z}/4\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

Indeed we have a cofibre sequence:

$$F \to \mathbb{F}_2 \xrightarrow{\mathrm{Sq}^1} \Sigma\mathbb{F}_2$$

which is determined by $F \to \mathbb{F}_2$. But Sq^1 is nontrivial, this shows $F = \mathbb{Z}/4\mathbb{Z}$.

Spectra have interanl Hom and tensor products $\otimes_{\mathbb{S}}$.

Remark. *$\otimes$ is associative and commutative. The associativity says there is a functor*

$$F : \mathbf{\Delta}^{\mathrm{op}} \to \mathrm{Cat}_\infty$$

satisfying the Segal conditions (the map $F([n]) \to F([1])^{\times n}$ is an equivalence). The commutativity comes from the the $\mathbb{E}_\infty$-structure.

If X is a space, E a spectrum, we defined $E^*(X)$ and $E_*(X)$. Note $E_*(\Delta^0) \simeq \pi_* E$.

Example. *Let E be the fibre of $\mathbb{F}_2 \xrightarrow{\mathrm{Sq}^8} \Sigma^8\mathbb{F}_2$. We have:*

$$E_*(\Delta^0) = \begin{cases} \mathbb{F}_2, & \text{if } * = 0, 2; \\ 0, & \text{otherwise.} \end{cases}$$

We also computed

$$E^*(\Delta^0) = \pi_{-*}(E) = \begin{cases} \mathbb{F}_2, & \text{if } * = 0, -7; \\ 0, & \text{otherwise.} \end{cases}$$

Suppose X is any space, we have

$$E^*X = \pi_{-*}\underline{\mathrm{Hom}}(\Sigma^\infty_+ X, E)$$

Also $\underline{\mathrm{Hom}}(\Sigma^\infty_+ X, E)$ is the fibre of $\underline{\mathrm{Hom}}(\Sigma^\infty_+ X; \mathbb{F}_2) \to \underline{\mathrm{Hom}}(\Sigma^\infty_+, \Sigma^8\mathbb{F}_2)$ There is a LES:

$$\cdots \to E^*X \to H^*(X; \mathbb{F}_2) \xrightarrow{\mathrm{Sq}^8} H^{*+8}(X; \mathbb{F}_2) \to E^{*+1}X \to \cdots$$

9.5 Spectral Sequence for Spectra

If E is a spectra, then E is the colimit

$$\cdots \to \tau_{\geq 1}E \to \tau_{\geq 0}E \to \tau_{\geq -1}E \to \cdots$$

This is the Whitehead tower. We have cofibre sequence: $\tau_{\geq i+1}E \to \tau_{\geq i}E \to \Sigma^i \pi_i E$.

E is the limit

$$\cdots \to \tau_{\leq 1}E \to \tau_{\leq 0}E \to \tau_{\leq -1}E \to \cdots$$

This is the Postinikov tower. We have cofibre sequence $\Sigma^i \pi_i E \to \tau_{\leq i}E \to \tau_{\leq i-1}E$.

If X is a space and E is a spectrum, we can write $E \otimes \Sigma^\infty_+ X$ as colimit of

$$(\cdots \to \tau_{\geq n}E \otimes \Sigma^\infty_+ X \to \tau_{\geq n-1}E \otimes \Sigma^\infty_+ X \to \cdots)$$

The fibre of the above map is

$$\Sigma^{n-1}\pi_{n-1}E \otimes \Sigma^\infty_+ X$$

which has homotopy

$$\pi_*(\Sigma^{n-1}\pi_{n-1}E \otimes \Sigma^\infty_+ X) = H_{*-n+1}(X; \pi_{n-1}E)$$

If E is any spectrum and X is a space, we have a spectral sequence

$$H_p(X, \pi_q E) \Rightarrow E_{p+q}(X)$$

This is called *Atiyah-Hirzebruch spectral sequence.*

More generally, suppose $F \to Y \to B$ is a fibre sequence of pointed spaces with B simply connected. We have a spectral sequence

$$H_p(B, E_q F) \Rightarrow E_{p+q}Y$$

Here is the construction. Given $F \to Y \to B$, we write $Y = \text{colim}_B F$. Hence $\Sigma^\infty_+ Y = \text{colim}_B \Sigma^\infty_+ F$. Tensoring with E:

$$\begin{aligned} E \otimes \Sigma^\infty_+ Y &= \text{colim}_B E \otimes \Sigma^\infty_+ F \\ &= \text{colim}_B \text{colim}_n((\tau_{\geq n}E) \otimes \Sigma^\infty_+ F) \\ &= \text{colim}_n \text{colim}_B((\tau_{\geq n}E) \otimes \Sigma^\infty_+ F) \end{aligned}$$

Remark. *There is a spectral sequence $H^p(B; E^q F)$, it may* **not** *converge to E^{p+q} of the total space. But we do have convergence:*

$$H^p(X; E^q(\Delta^0) = \pi_{-q}(E)) \Rightarrow E^{p+q}(X)$$

if X has finitely many homology groups, or E has finitely many π_* groups.

10 Higher Algebra

10.1 Homotopy Ring

Definition 10.1. A homotopy ring is the data of a spectrum E with a map $E \otimes_{\mathbb{S}} E \to E$.

If E is a homotopy ring and X is a space, then given two maps:

$$\Sigma^\infty_+ X \xrightarrow{f} \Sigma^i E$$

and

$$\Sigma^\infty_+ X \xrightarrow{g} \Sigma^j E$$

we get a map:

$$\Sigma^\infty_+ X \xrightarrow{\Delta} \Sigma^\infty_+ (X \times X) \simeq \Sigma^\infty_+ X \otimes \Sigma^\infty_+ X \to \Sigma^i E \otimes_{\mathbb{S}} \Sigma^j E \xrightarrow{m} \Sigma^{i+j} E$$

This gives $E^* X$ a ring structure: has a multiplication which distributes over addition (need not be associative, commutative, or unital).

Definition 10.2. We say E is homotopy unital if there is a map $\mathbb{S} \to E$ such that

$$E \otimes \mathbb{S} \to E \otimes E \to E$$

and

$$\mathbb{S} \otimes E \to E \otimes E \to E$$

are homotopic to 1_E.

Definition 10.3. A homotopy associative ring is a homotopy ring with the property that

$$\begin{array}{ccc} E \otimes E \otimes E & \xrightarrow{m \otimes 1} & E \otimes E \\ {\scriptstyle 1 \otimes m}\downarrow & & \downarrow{\scriptstyle m} \\ E \otimes E & \xrightarrow{m} & E \end{array}$$

commutes up to homotopy. An $\mathbb{E}_1$-ring comes with the data of a chosen homotopy filling the square, and all other associators.

If E is $\mathbb{E}_1$, then $\Omega^\infty E$ is an $\mathbb{E}_1$ space in two ways: $(+, \cdot)$. We also have $\mathbb{E}_\infty$-rings E, such that $\Omega^\infty E$ has two different $\mathbb{E}_\infty$-space structures.

Example. *Any discrete associative ring is a $\mathbb{E}_1$-ring. Any discrete commutative ring is an $\mathbb{E}_\infty$-ring.*

If R is an $\mathbb{E}_1$-ring, then $\pi_* R = R^{-*}(\Delta^0)$ is an associative ring. If R is an $\mathbb{E}_\infty$-ring, then $\pi_* R$ is a graded commutative ring.

Remark. *The groupoid of finite sets (with bijection) is an $\mathbb{E}_\infty$ space. The group completion of it is $\Omega^\infty \mathbb{S}$.*

Remark. $\mathbb{S}$ *is the initial* $\mathbb{E}_\infty$*-ring. We recall:*

$$\pi_0\mathbb{S} = \mathbb{Z} \quad \pi_1\mathbb{S} = \mathbb{Z}/2\mathbb{Z}\{\eta\} \quad \pi_2\mathbb{S} = \mathbb{Z}/2\mathbb{Z}\{\eta^2\} \quad \eta^4 = 0 \quad \eta^3 = 8\nu \cdots$$

If $x \in \pi_{2i+1}\mathbb{S}$*, then* $x^2 = -x^2 \implies x^2$ *is a 2-torsion.*

Example. *If* E *is Eilenberg-Maclane for a commutative ring, it is* $\mathbb{E}_\infty$*.*

Example. *Recall we have a* $\mathbb{E}_\infty$*-ring:*

$$KU = (BU \times \mathbb{Z}, U, BU \times \mathbb{Z}, U, \cdots)$$

$\tau_{\geq 0}KU$ *is the group completion of* $\mathbb{C}$*-vector spaces under* $\oplus$ *and* $\otimes$*. Hence:*

$$\pi_* KU = \mathbb{Z}[\beta^{\pm}]$$

where $\deg(\beta) = 2$*.*

Example. *Suppose we want to compute:*

$$KU^*(\mathbb{CP}^n)$$

We have spectral sequence:

$$H^p(\mathbb{CP}^n; KU^q(\Delta^0)) \simeq \mathbb{Z}[\beta^{\pm}][t]/(t^{n+1}) \Rightarrow KU^{p+q}(\mathbb{CP}^n)$$

with $\deg f = (2,0), \deg \beta = (0,-2)$*. By parity reason, there are no differentials.*

Definition 10.4. An even $\mathbb{E}_\infty$-ring is an $\mathbb{E}_\infty$ ring such that $\pi_* R \simeq 0$ for all odd degrees.

Example. *Discrete commutative rings, and* KU *are even* $\mathbb{E}_\infty$ *ring, but* $\mathbb{S}$ *is not.*

Proposition 10.5. *If* E *is an even* $\mathbb{E}_\infty$*-ring, then*

$$E^*(\mathbb{CP}^n) \simeq (\pi_{-*}E)[t]/t^{n+1}$$

as a ring.

Proof. Follows by parity in AH Spectral Sequence. □

Remark. *Suppose*

$$\cdots \to A_1 \to A_0 \to A_{-1} \to \cdots$$

is a sequence of spectra. Then

$$\pi_p(\mathrm{colim} A_i) = \mathrm{colim} \pi_p(A_i)$$

This is in fact true for any filtered colimits.

Proposition 10.6 (Milnor exact sequence)**.** *The natural map*

$$\pi_q(\lim A_i) \to \lim \pi_q A_i$$

fits into:

$$0 \to \lim{}^1 \pi_{q+1} A_i \to \pi_q(\lim A_i) \to \lim \pi_q A_i \to 0$$

Remark. *Mittag-Lefschetz condition says* $\lim^1$ *will often vanish, for example whenever the map*

$$\pi_q A_i \to \pi_q A_{i_1}$$

are surjective for $i \gg 0$.

Proof. (Sketched). Note:

$$\lim A_i = \text{pullback}\Big(\begin{array}{ccc} & & \prod A_i \\ & & \downarrow{\scriptstyle (1,1)} \\ \prod A_i & \xrightarrow[(1,\text{shift})]{} & \prod A_i \oplus A_i \end{array} \Big)$$

and

$$\pi_q(\prod A_i) = \prod \pi_q(A_i)$$

and we then use LES for π_* of a pullback. □

Suppose E is an even $\mathbb{E}_\infty$-ring, then:

$$\begin{aligned} E^*\mathbb{CP}^\infty &= \pi_0 \underline{\text{Hom}}(\mathbb{CP}^\infty, \Sigma^* E) \\ &= \pi_0 \underline{\text{Hom}}(\text{colim}\mathbb{CP}^n, \Sigma^* E) \\ &= \pi_0 \lim \underline{\text{Hom}}(\mathbb{CP}^n, \Sigma^* E) \\ &= \lim \pi_0 \underline{\text{Hom}}(\mathbb{CP}^n, \Sigma^* E) \\ &= (\pi_{-*} E)[[t]] \end{aligned}$$

Remark. $\mathbb{Z}^*\mathbb{CP}^\infty$ *is a* graded *power series ring:*

$$\mathbb{Z}^*\mathbb{CP}^\infty \simeq \mathbb{Z}[[t]] \simeq \mathbb{Z}[t]$$

for t *in degree 2.*

Example.

$$KU^0(\mathbb{CP}^\infty) \simeq \mathbb{Z}[[\beta t]]$$

is an honest power series ring. We write:

$$KU^*(\mathbb{CP}^\infty) = \mathbb{Z}[\beta^{\pm}][[t]] = \mathbb{Z}[\beta^{\pm}][[\beta t]]$$

Suppose X is a space with a line bundel $\mathcal{L} : X \to \mathbb{CP}^\infty$. Then $c_1(\mathcal{L}) \in E^2 X$ is the image of $t \in E^2(\mathbb{CP}^\infty)$. If $\mathcal{L}_1, \mathcal{L}_2$ are line bundles on X ,there will be a formula for $c_1(\mathcal{L}_1 \otimes \mathcal{L}_2)$ in terms of $c_1(\mathcal{L}_1)$ and $c_1(\mathcal{L}_2)$. THis formula comes from calculation of:

$$E^2(\mathbb{CP}^\infty \times \mathbb{CP}^\infty \xrightarrow{m} \mathbb{CP}^\infty)$$

Example. *On homework, we saw if* $E = \mathbb{Z}$*, then:*

$$c_1(\mathcal{L}_1 \otimes \mathcal{L}_2) = c_1(\mathcal{L}_1) + c_1(\mathcal{L}_2)$$

In fact, if $E = KU$

$$c_1(\mathcal{L}_1 \otimes \mathcal{L}_2) = c_1(\mathcal{L}_1) + c_1(\mathcal{L}_2) + \beta c_1(\mathcal{L}_1) c_1(\mathcal{L}_2)$$

In general, the formula is a formal group.

Remark. *If* R *is an* $\mathbb{E}_1$*-ring, there is a category of left* R*-modules. We know* $D(\mathbb{Z})$ *is the category of* $\mathbb{Z}$*-modules,* Spectra *is the category of* $\mathbb{S}$*-modules.*

10.2 Modules

If R is an $\mathbb{E}_1$-ring, then (left) R-modules is a category that can be defined. Very roughly, a left module M is a spectrum M with an action map $R \otimes_{\mathbb{S}} M \to M$, with a choice of homotopy filling the obvious associativity diagram.

Remark. Spectra $= \mathbb{S}$-*modules.* $D(\mathbb{Z}) = \mathbb{Z}$-*modules.*

Remark. *If $\mathcal{C}$ is a stable category, and c an object in $\mathcal{C}$. Then: $\underline{\mathrm{Hom}}_{\mathcal{C}}(c,c)$ is naturally an $\mathbb{E}_1$-ring. An R-module structure on a spectrum M is an $\mathbb{E}_1$-ring map $R \to \mathrm{Hom}_{\mathrm{Spectra}}(M,M)$.*

Remark. *$R = \underline{\mathrm{Hom}}_{R\text{-mod}}(R,R)$ as an $\mathbb{E}_1$-ring.*

Theorem 10.7 (Schwede-Shipley)**.** *Suppose $\mathcal{C}$ is a presentable stable category with a compact generator c. Then*

$$\mathcal{C} = \underline{\mathrm{Hom}}_{\mathcal{C}}(c,c)\text{-mod}$$

$D(\mathbb{Z})$ is such a category, with a compact generator $c = (\cdots \to 0 \to \mathbb{Z} \to 0 \to \cdots)$ concentrated in degree 0. One computes:

$$\pi_i(\underline{\mathrm{Hom}}_{D(\mathbb{Z})}(c,c)) = \begin{cases} \mathbb{Z}, & \text{if } i = 0; \\ 0, & \text{otherwise.} \end{cases}$$

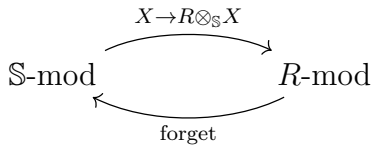
Remark. *π_* of a $\mathbb{Z}$-module is homology of a chain complex. Suppose M is a $\mathbb{Z}$-module, then:*

$$\pi_0 \underline{\mathrm{Hom}}_{\mathbb{Z}\text{-mod}}(c[-n], M) = H_n(M)$$

Hence we can compute:

$$\begin{aligned} H_n(M) &= \pi_0 \underline{\mathrm{Hom}}_{\mathbb{Z}-\mathrm{mod}}(\Sigma^n \mathbb{Z}, M) \\ &= \pi_0 \underline{\mathrm{Hom}}_{\mathbb{Z}-\mathrm{mod}}(\mathbb{Z} \otimes_{\mathbb{S}} \Sigma^n \mathbb{S}, M) \\ &= \pi_0 \underline{\mathrm{Hom}}_{\mathrm{Spectra}}(\Sigma^n \mathbb{S}, \mathrm{forget}(M)) = \pi_n(M) \end{aligned}$$

There's something to check: if R is any $\mathbb{E}_1$-ring, we have an adjunction:



Remark. *If R is $\mathbb{E}_\infty$, R-modules has a (symmetric monoidal) tensor product $\otimes_R$. Therefore, R-modules are enriched over themselves. In fact, if R is $\mathbb{E}_n$, then R-modules is $\mathbb{E}_{n-1}$.*

Remark. *If R is discrete commutative ring, with M, N discrete modules. Then:*

$$\mathrm{Tor}_i^R(M,N) = \pi_i(M \otimes_R N)$$

$$\mathrm{Ext}_R^i(M,N) = \pi_{-i} \underline{\mathrm{Hom}}_{R-\mathrm{mod}}(M,N)$$

Remark. *The functor from chain complex of $\mathbb{Z}$-mod to the chain complexes of $\mathbb{Z}$-modules with quasi-iso inverted preserves $\otimes$ for levelwise free chain complexes.*

Definition 10.8. If R is an $\mathbb{E}_1$-ring, then a free left R-module is any R-module isomorphic to $\bigoplus_{i\in I} \Sigma^{n_i} R$

Definition 10.9. An $\mathbb{E}_1$-ring R is a (skew) field if every R-module is free.

Remark. *If R is an $\mathbb{E}_1$-ring, and M is a module, then $\pi_* M$ is a graded $\pi_* R$-module.*

Proposition 10.10. *If k is an Eilenberg-Maclane (skew) field. Then k is a (skew) field.*

Proof. If M is a k-module in the sense of spectra. Then $\pi_* M$ is a free graded $\pi_*(k)$-module. Hence we can "choose a basis":

$$\pi_* M = \bigoplus_{i\in I} \pi_*(k)[n_i]$$

This gives each $i \in I$ a map of spectra: $S^{n_i} \to M$. Collectively, we have a map of spectra

$$\bigoplus_{i\in I} S^{n_i} \to M$$

which has image a basis for $\pi_* M$ as a graded $\pi_* k$-vector space. This corresponds to

$$k \otimes_{\mathbb{S}} (\bigoplus_{i\in I} S^{n_i}) \to M \iff \bigoplus \Sigma^{n_i} k \to M$$

which is an isomorphism on π_*. We thus conclude by Whitehead. □

Remark. *Suppose R is an $\mathbb{E}_1$-ring such that $\pi_* R$ is a graded skew field. Then R is a skew field.*

Example. *Let $R = KU/p$ is an $\mathbb{E}_1$-ring with homotopy $\pi_* R \simeq \mathbb{F}_p[\beta^{\pm}]$ is a skew field.*

Definition 10.11. Two skew fields R_1, R_2 have the same characteristic if $R_1 \otimes_{\mathbb{S}} R_2 \neq 0$.

Remark. *To check the above is an equivalence relation, we need to realize $R_1 \otimes_{\mathbb{S}} R_2 \neq 0$ is botha a free R_1-module and free R_2-module.*

Remark. *Every discrete field the characteristic of either $\mathbb{F}_p$ for some prime p or $\mathbb{Q}$.*

Lemma 10.12. *If R is a field, then $R \otimes_{\mathbb{Z}} R \neq 0$*

Proof. The composition:

$$\mathbb{S} \otimes_{\mathbb{S}} R \to R \otimes_{\mathbb{S}} R \xrightarrow{m} R$$

is the identity map. □

Proposition 10.13. *The skew field KU/p has a different characteristic.*

Proof. Let $BC_p = K(\mathbb{F}_p, 1)$. Note $(KU/p)^*(BC_p)$ is a module over $(KU/p)^*(\Delta^0)$ of dimension p. But the former is just a mod p-representation ring of C_p. But if we consider $\mathbb{F}_p^*(BC_p)$ is a $\mathbb{F}_p$ vector space of infinite rank, $\mathbb{F}_\ell^*(BC_p)$ has rank 1, $\mathbb{Q}^*BC_p$ has rank 1. □

Remark. *Every field has either characteristic $\mathbb{Q}$, or $\pi_0 R$ has characteristic p for some prime p. In the later case, $R^*(BC_p)$ will be free of rank p^h for some h over $R^*(\Delta^0)$. The integer h determines the characteristic.*

Definition 10.14. If R is an $\mathbb{E}_\infty$-ring, and $x \in \pi_p R$, viewed as an R-module map $x : \Sigma^p R \to R$. Then

$$R[x^{-1}] = \operatorname{colim}(R \xrightarrow{x} \Sigma^{-p} R \xrightarrow{x} \Sigma^{-2p} R \to \cdots)$$

Proposition 10.15. $\pi_*(R[x^{-1}]) = (\pi_* R)[x^{-1}]$

Theorem 10.16. *$R[x^{-1}]$ is an $\mathbb{E}_\infty$-ring in a natural way.*

If R is an $\mathbb{E}_\infty$-ring, and $x \in \pi_p R$, we can form:

$$R/x = \operatorname{cofibre}(\Sigma^p R \xrightarrow{x} R) = \operatorname{colim}_{R\text{-mod}}\left(\begin{array}{ccc} \Sigma^p R & \xrightarrow{x} & R \\ \downarrow & & \\ 0 & & \end{array} \right)$$

We have a long exact sequence:

$$\cdots \to \pi_* R \xrightarrow{x} \pi_{*+p} R \to \pi_{*+p} R/x \to \pi_{*+p-1} R \xrightarrow{x} \pi_{*+2p-1} R \to \cdots$$

Remark. *$\mathbb{S}^2/2$ is not a ring. Indeed $\pi_*(\mathbb{S}/2)$ has an element of order 4. Also KU/p is $\mathbb{E}_1$ but not $\mathbb{E}_2$.*

If R is an $\mathbb{E}_\infty$-ring and $x \in \pi_p R$ is an element.

Definition 10.17. If M is any R-module, then:

$$R/x \otimes_R M = M/x \coloneqq \operatorname{cofibre}(\Sigma^p R \times_R M \xrightarrow{x \otimes 1} R \otimes_R M) = \operatorname{cofibre}(\Sigma^p M \xrightarrow{x} M)$$

$$R[x^{-1}] \otimes_R M = M[x^{-1}] = \operatorname{colim}(M \xrightarrow{x} \Sigma^{-p} M \xrightarrow{x} \Sigma^{-2p} M \to \cdots)$$

In fact $R[x^{-1}]$ remains an $\mathbb{E}_\infty$-ring.

$$\widehat{M_x} \coloneqq \lim(\cdots \to M/x^2 \to M/x)$$

is a R-module

Remark. *$\widehat{R_x}$ is an $\mathbb{E}_\infty$-ring.*

Remark. *$\widehat{M_x} \neq \widehat{R}_x \otimes_R M$ in general (this already fails in ordinary algebra, because $\otimes$ does not commute with limits). However, this is true if we can write M as some colimit of R.*

Example.

$$\begin{aligned}\underline{\mathrm{Hom}}_{\mathbb{Z}}(\mathbb{F}_p, \mathbb{Z}) &= \underline{\mathrm{Hom}}_{\mathbb{Z}}(\mathrm{cofibre}(\mathbb{Z} \xrightarrow{p} \mathbb{Z}), \mathbb{Z}) \\ &= \mathrm{fibre}(\underline{\mathrm{Hom}}(\mathbb{Z}, \mathbb{Z}) \xrightarrow{p} \underline{\mathrm{Hom}}(\mathbb{Z}, \mathbb{Z})) \\ &= \mathrm{fibre}(\mathbb{Z} \xrightarrow{p} \mathbb{Z}) = \Sigma^{-1}\mathbb{F}_p\end{aligned}$$

Thus there exists a nontrivial map

$$S^{-1} \to \underline{\mathrm{Hom}}_{\mathbb{Z}}(\mathbb{F}_p, \mathbb{Z}) \iff \Sigma^{-1}\mathbb{F}_p \to \mathbb{Z}$$

This is precisely the map appearing in the fibre sequence:

$$\Sigma^{-1}\mathbb{F}_p \to \mathbb{Z} \xrightarrow{p} \mathbb{Z}$$

There is a $\mathbb{Z}$-module map:

$$\Sigma^{-1}\mathbb{F}_p \to \mathbb{Z} \to \mathbb{F}_p$$

This is called the **Bockstein** map. There is a fibre sequence:

$$\mathbb{Z}/p^2 \to \mathbb{F}_p \xrightarrow{\mathrm{Sq}^1} \Sigma\mathbb{F}_p$$

Remark. $\mathbb{F}_2 \xrightarrow{\mathrm{Sq}^3} \Sigma^3\mathbb{F}_2$ *is an* $\mathbb{S}$*-module map. This is not a* $\mathbb{Z}$*-module map since*

$$\mathrm{Ext}^3_{\mathbb{Z}}(\mathbb{F}_2, \mathbb{F}_2) = 0$$

Example. *Let* $\mathbb{S}/2 = \mathrm{cofibre}(\mathbb{S} \xrightarrow{2} \mathbb{S})$*. We can compute:*

$$H^*(\mathbb{S}/2, \mathbb{F}_2) = \begin{cases} \mathbb{F}_2, & \text{if } * = 0, 1; \\ 0, & \text{otherwise.} \end{cases}$$

Note we have a fibre sequence in spectra:

$$\Sigma^{-1}\mathbb{S}/2 \to \mathbb{S} \xrightarrow{2} \mathbb{S}$$

tensoring with $\mathbb{Z}$*, we get:*

$$\Sigma^{-1}\mathbb{F}_2 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z}$$

In fact the first map compose with $\mathbb{Z} \to \mathbb{F}_2$ *is the* Sq^1*.*

Let X be a spectrum.

Proposition 10.18. *Let* k *be a skew field, then:*

$$H_*(X, k) \simeq H^*(X, k)$$

Proof. Note that $k \otimes_{\mathbb{S}} X$ is a (free) k-module, therefore we can write:

$$k \otimes_{\mathbb{S}} X = \bigoplus_{i \in I} \Sigma^{n_i} k$$

Hence:

$$H_*(X, k) = \pi_*(k \otimes_{\mathbb{S}} X) = \bigoplus_{i : n_i = *} k$$

We also see:

$$\underline{\mathrm{Hom}}_{\mathbb{S}}(X, k) \simeq \underline{\mathrm{Hom}}_k(k \otimes_{\mathbb{S}} X, k) = \bigoplus_{i \in I} \underline{\mathrm{Hom}}_k(k, \Sigma^{-n_i} k) = \bigoplus_{i \in I} \Sigma^{-n_i} k$$

Hence:

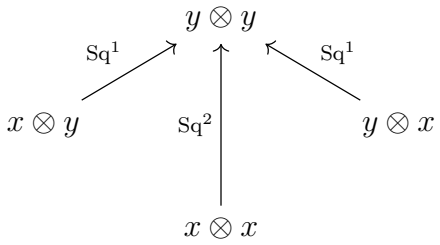
$$H^*(X, k) = \pi_0 \Sigma^* \underline{\mathrm{Hom}}_{\mathbb{S}}(X, k) = \bigoplus_{i : n_i = *} k$$

□

Theorem 10.19 (Kunneth)**.**

$$\mathbb{F}_2 \otimes_{\mathbb{S}} X \otimes_{\mathbb{S}} Y = (\mathbb{F}_2 \otimes_{\mathbb{S}} X) \otimes_{\mathbb{F}_2} (\mathbb{F}_2 \otimes_{\mathbb{S}} Y)$$

Example. *$H^*(\mathbb{S}/2 \otimes_{\mathbb{S}} \mathbb{S}/2; \mathbb{F}_2)$ is 4-dimensional.*



Remark. *If $\mathbb{S}/2$ were a unital ring, then*

$$\mathbb{S}/2 = \mathbb{S}/2 \otimes \mathbb{S} \to \mathbb{S}/2 \otimes \mathbb{S}/2 \xrightarrow{m} \mathbb{S}/2$$

would make $H^*(\mathbb{S}/2)$ be a retract of $H^*(\mathbb{S}/2 \otimes \mathbb{S}/2)$. But this is not the case.

Proposition 10.20. *Let R be a skew field and p is a prime. Then either $p = 0$ in $\pi_0(R)$, or p acts invertibly on $\pi_0(R)$.*

Proof. We only prove the statement for $p > 2$. Assume $p \neq 0 \in \pi_0(R)$. For p odd prime, we have $\mathbb{S}/p$ is a retract of $\mathbb{S}/p \otimes \mathbb{S}/p$ (even though in fact $\mathbb{S}/p$ is not $\mathbb{E}_1$ ring). Namely, $\mathbb{S}/p$ admits a unital multiplication but that multiplication is not coherently associative. Once we know this, from the retract:

$$\mathbb{S}/p \to \mathbb{S}/p \otimes \mathbb{S}/p \to \mathbb{S}/p$$

tensoring over R we have:

$$R/p \to R/p \otimes_R R/p \to R/p$$

To show p acts invertibly, it suffices to show $R/p = 0$. Suppose not, since R is a skew field, so we can write $R/p = \bigoplus R[n_i]$, therefore the first map becomes:

$$\bigoplus R[n_i] \to \bigoplus R/p[n_i]$$

This map admits a retract, therefore $R \to R/p$ admits a retract, hence $\pi_* R \to \pi_* R/p$ is injective for each index. But we have the exact sequence:

$$\pi_0(R) \xrightarrow{p} \pi_0(R) \to \pi_0(R/p)$$

This means $p = 0$, contradiction. □

10.3 Dual Steenrod Algebra

Recall

$$\mathcal{A} \coloneqq \mathbb{F}_2^* \mathbb{F}_2 = \pi_{-*} \underline{\mathrm{Hom}}(\mathbb{F}_2, \mathbb{F}_2) = \mathbb{F}_2\{\mathrm{Sq}^1, \mathrm{Sq}^2, \cdots\}/\{\text{Adam relations}\}$$

It's natural for us to define:

$$\mathcal{A}_* \coloneqq \mathbb{F}_{2*} \mathbb{F}_2 = \pi_*(\mathbb{F}_2 \otimes_{\mathbb{S}} \mathbb{F}_2)$$

is a commutative ring, because tensor product of $\mathbb{E}_\infty$-rings is $\mathbb{E}_\infty$. This is called the **dual Steenrod algebra**.

Note:

$$\begin{aligned} \mathcal{A} &= \pi_{-*} \underline{\mathrm{Hom}}_{\mathbb{S}}(\mathbb{F}_2, \mathbb{F}_2) \\ &= \pi_{-*} \underline{\mathrm{Hom}}_{\mathbb{F}_2}(\mathbb{F}_2 \otimes_{\mathbb{S}} \mathbb{F}_2, \mathbb{F}_2) \end{aligned}$$

Hence $\mathcal{A}$ is indeed the $\mathbb{F}_2$-linear dual of $\mathcal{A}_*$.

Remark. *The construction of EM spaces using Dold-Kan correspondence implies that they have finitely many nondegenerate simplices in each dimension. Hence $H_*(-; \mathbb{F}_2)$ is finite-dimensional in each degree. So $\pi_*(\mathbb{F}_2 \otimes_{\mathbb{S}} \mathbb{F}_2)$ is finite in each degree. As a consequence, we see $\mathcal{A}_*$ is also the dual of $\mathcal{A}$.*

The multiplication map

$$\mathcal{A}_* \otimes_{\mathbb{F}_2} \mathcal{A}_* \to \mathcal{A}_*$$

gives on dual a map "comultiplication"

$$\mathcal{A} \to \mathcal{A} \otimes_{\mathbb{F}_2} \mathcal{A}$$

This is a graded associative ring map. In fact, Milnor proved this is the map:

$$\mathrm{Sq}^k \mapsto \sum_{i+j=k} \mathrm{Sq}^i \otimes \mathrm{Sq}^j$$

Theorem 10.21 (Milnor)**.**

$$\pi_*(\mathbb{F}_2 \otimes_{\mathbb{S}} \mathbb{F}_2) = \mathbb{F}_2[\xi_1, \xi_2, \cdots]$$

where $|\xi_i| = 2^i - 1$.

Remark. *ξ_1 is dual to* Sq^1*, ξ_2 is dual to* $[\mathrm{Sq}^2, \mathrm{Sq}^1]$*, ξ_3 is dual to* $[\mathrm{Sq}^4, [\mathrm{Sq}^2, \mathrm{Sq}^1]], \cdots$*. We may denote Q_i as the dual of ξ_i.*

Remark. *$KU/2$ is a spectrum* $\mathrm{cofibre}(KU \xrightarrow{2} KU)$*. Note:*

$$\pi_*(\tau_{\leq 2}\tau_{\geq 2} KU/2) = \begin{cases} \mathbb{F}_2, & \text{if } * = 0, 2; \\ 0, & \text{otherwise.} \end{cases}$$

In fact: $\tau_{\leq 2}\tau_{\geq 2}KU/2 = \mathrm{fibre}(\mathbb{F}_2 \xrightarrow{Q_2} \Sigma^3 \mathbb{F}_2)$

Remark. $KU = \Sigma^\infty_+ \mathbb{CP}^\infty[\beta^{-1}]$.

Remark. *We have*

$$\mathrm{sAb} \simeq \{\text{chain complexes concentrated in nonnegative degrees}\} \to D(\mathbb{Z})_{\geq 0}$$

This is the functor $(\Delta^{\mathrm{op}} \to \mathrm{Ab}) \mapsto \mathrm{colim}(\Delta^{\mathrm{op}} \to \mathrm{Ab} \to D(\mathbb{Z}))$.

One can define cosimplicial abelian group as a functor $\mathbf{\Delta} \to \mathrm{Ab}$*. We can obtain an object of $D(\mathbb{Z})$ as limit* $\lim(\mathbf{\Delta} \to \mathrm{Ab} \to D(\mathbb{Z}))$.

Suppose $A \to B$ is a map of $\mathbb{E}_\infty$-rings. Consider $\mathcal{I} = \mathrm{fib}(A \to B)$, then we get:

$$\cdots \to \mathcal{I} \otimes_A \mathcal{I} \otimes_A \mathcal{I} \to \mathcal{I} \otimes_A \mathcal{I} \to \mathcal{I} \to A$$

This is a filtration of A. There is a spectral sequence trying to converge to $\pi_* A$, with the E_1-page:

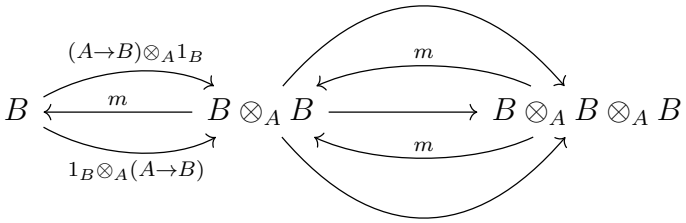
$$\bigoplus \pi_p \mathrm{cofib}(\mathcal{I}^{\otimes_A q+1} \to \mathcal{I}^{\otimes_A q})$$

This is called the **Adam Spectral Sequence**.

Example. *If $A = \mathbb{S}$ and $B = \mathbb{F}_2$, then*

$$\pi_*(\mathrm{fib}(\mathbb{S} \to \mathbb{F}_2) \otimes_{\mathbb{S}} \mathbb{F}_2) = \pi_* \mathrm{fib}(\mathbb{F}_2 \to \mathbb{F}_2 \otimes_{\mathbb{S}} \mathbb{F}_2)$$

Another way to think about the Adams spectral sequence for $A \to B$ is the following. Given a map of $\mathbb{E}_\infty$-rings $A \to B$, can form a diagram of a cosimplicial A-module:



The limit of this diagram is "trying" to be A. The limit is called the B-completion of A, denoted as A_B.

Example. *If $A \to B$ is such that $\mathcal{I}$ has π_* concentrated in $* \geq 1$. Then $A \to A_B$ is an equivalence. For example, this is the case for $\mathbb{S} \to \mathbb{Z}$.*

Example. $\mathbb{S}_{\mathbb{F}_2} = \widehat{\mathbb{S}_2}$

Remark. $\mathbb{S}_{KU}$ *is completely understood.* $\widehat{(\mathbb{S}_{KU})_p} \to \widehat{KU_p}$ *is Galois with Galois group* $\widehat{\mathbb{Z}_p}$

Remark. *Given $A \to B$, we can define:*

$$\mathrm{fil}^n = \lim_{[k] \in \mathbf{\Delta}} \tau_{\geq n}(B^{\otimes_A k+1})$$

Then:

$$0 \to \cdots \to \mathrm{fil}^1 \to \mathrm{fil}^0 \to \mathrm{fil}^{-1} \to \cdots \to A_B$$

Note

$$\mathrm{gr}^n = \mathrm{cofib}(\mathrm{fil}^{n+1} \to \mathrm{fil}^n)$$

is Σ^n of a $\mathbb{Z}$-module represented by the chain complex:

$$\cdots \to \pi_n B \xrightarrow{\eta_L - \eta_R} \pi_n(B \otimes_A B) \to \pi_n(B^{\otimes_A 3}) \to \cdots$$

When $B = \mathbb{F}_2$ and $A = \mathbb{S}$, we get a spectral sequence starts with the p-th homology of

$$\pi_q \mathbb{F}_2 \xrightarrow{\eta_L - \eta_R} \pi_q(\mathbb{F}_2 \otimes_{\mathbb{S}} \mathbb{F}_2) \to \pi_q(\mathbb{F}_2 \otimes \mathbb{F}_2 \otimes \mathbb{F}_2) \to \cdots$$

This is computing

$$\mathrm{Ext}_{\mathcal{A}}(\mathbb{F}_2, \mathbb{F}_2) \Rightarrow \pi_* \widehat{\mathbb{S}_2}$$

We hope to find a diagram

$$\mathbb{S} \to E \to \mathbb{F}_2$$

such that we can use descent along $E \to \mathbb{F}_2$ to understand $\pi_* E$, and we can use descent along $\mathbb{S} \to E$ to understand $\pi_* \mathbb{S}$. The E we will use is MU.

10.4 A Study on MU

Informally, MU is the E_∞ ring (under disjoint union and product):

- Objects: 0-dimensional complex manifolds
- Morphisms: 1-dimensional complex bordisms
- etc.

It will turn out that $\pi_*(MU^{\otimes n})$ are concentrated in even degrees. In particular, MU is an even $\mathbb{E}_\infty$-ring.

Recall if E is an E_∞-ring, then

$$E^* \mathbb{CP}^n \simeq (\pi_{-*} E)[t]/t^n$$

$$\implies E^* \mathbb{CP}^\infty \simeq (\pi_{-*} E)[[t]]$$

Definition 10.22. A **complex orientation** of an even E_∞-ring is a choice of isomorphism:

$$E^*\mathbb{CP}^\infty \simeq (\pi_{-*}E)[[t]]$$

This is a map:

$$\Sigma^\infty\mathbb{CP}^\infty \xrightarrow{t} \Sigma^2 E$$

such that the composite:

$$S^2 = \Sigma^\infty\mathbb{CP}^1 \to \Sigma^\infty\mathbb{CP}^\infty \to \Sigma^2 E$$

is the unit of the E_∞-ring E.

Note the AH Spectral Sequence collapses also for $\mathbb{CP}^i \times \mathbb{CP}^j$. If E is a complex oriented even E_∞-ring, there are canonical isomorphisms:

$$E^*\mathbb{CP}^\infty \simeq (\pi_{-*}[[t]])$$

and

$$E^*(\mathbb{CP}^\infty \times \mathbb{CP}^\infty) = (\pi_{-*}E)[[x, y]]$$

The multiplication map $\mathbb{CP}^\infty \times \mathbb{CP}^\infty \to \mathbb{CP}^\infty$ gives a map:

$$E^*\mathbb{CP}^\infty \to E^*\mathbb{CP}^\infty \times \mathbb{CP}^\infty$$

$$t \mapsto f(x, y) \in \pi_{-*}E[[x, y]]$$

Example. *If $E = \mathbb{Z}$, we proved on Homework that $f(x, y) = x + y$*

Example. *If $E = KU$, then there exists a $\mathbb{C}$-orientation of KU such that $f(x, y) = x + y + \beta xy$*

In general:

- $f(x, y) = x + y+$ higher order terms (multiplication on $\mathbb{CP}^\infty$ is unital)
- $f(x, y) = f(y, x)$ (commutative)
- $f(x, f(y, z)) = f(f(x, y), z)$ (associativity)

Remark. *Suppose R is a classical commutative ring. Then*

$$\widehat{\mathbb{A}}^1_R = \text{colimSpec } R[t]/t^n$$

A power series satsifying with coefficients in R satisfying the three conditions above is the same data as a commutative and associative addition law:

$$t : \widehat{\mathbb{A}}^1_R \times \widehat{\mathbb{A}}^1_R \to \widehat{\mathbb{A}}^1_R$$

This is a **Formal Group Law**.

Remark. *A formal group is a group objectin in formal schemes, which is (non-canonically) isomorphic to $\widehat{\mathbb{A}}^1_R$ with some group structure. Any even $\mathbb{E}_\infty$-ring E has a canonical formal group over* Spec $\pi_* E$. *A choice of complex orientation gives a formal group law.*

Remark. *Given a 1-dim algebraic group, its completion at the origin is a fomral group. For example,* $\mathbb{G}_a = \text{Spec } \mathbb{Z}[t]$ *goes to* $\mathbb{Z}^*\mathbb{CP}^\infty$*;* $\mathbb{G}_m = \text{Spec } \mathbb{Z}[t^\pm][\beta^\pm]$ *goes to* $KU^*\mathbb{CP}^\infty$*.*

Theorem 10.23 (Hopkins-Miller,Lurie)**.** *There is a functor from a* 1*-category* $\mathcal{C}$ *to even* $\mathbb{E}_\infty$*-rings.*

The category $\mathcal{C}$ consists of pair $(A, \mathcal{E})$, where $\mathcal{E}$ is an elliptic curve over Spec A (versal defomration over every geometric point). This functor send $(A, \mathcal{E})$ maps to even $\mathbb{E}_\infty$-ring R with $\pi_* R = A[\beta^\pm]$ and the formal group is $\widehat{\mathcal{E}}$. $\lim_{\mathcal{C}}$ of this funcgor is the $\mathbb{E}_\infty$-ring is the category of topological modular forms.

Recall MU is a spectrum such that $\pi_n(MU)$ is the collection of complex smooth projective varieties of underlying real dimension n up to algebraic bordism. This is an even $\mathbb{E}_\infty$-ring, with disjoint union and $\times$.

Remark. *A manifold* M *has "stably almost complex structure" when equipped with a commuting triangle:*

$$\begin{array}{ccccc} & & & & BU \\ & & & & \downarrow \\ M & \xrightarrow[TM]{} & BO(n) & \longrightarrow & BO \end{array}$$

in Spaces*.* MU *is bordism of manifolds with such structure.*

Recall there is a $\mathbb{E}_1$-space map:

$$U(n) \to \text{Aut}(S^{2n})$$

This gives

$$BU(n) \to B\text{Aut}(S^{2n}) \implies BU \to B\text{GL}_1\mathbb{S}$$

Recall $B\text{GL}_1\mathbb{S}$ is the subcategory of spectra with one object $\mathbb{S}$ and morphisms the automorphism of $\mathbb{S}$.

Definition 10.24. $MU = \text{colim}(BU \to B\text{GL}_1\mathbb{S} \to \text{Spectra})$

Theorem 10.25 (Thom)**.** *This has homotopy groups related to bordism of stably almost complex manifolds.*

Remark. *If* V *is a complex vectord space, let* S^V *denote its one-point compactification.*

$$\Sigma^\infty S^{V \oplus W} = \Sigma^\infty S^V \otimes \Sigma^\infty S^W$$

So $BU \to B\text{GL}_1\mathbb{S}$ *is an* $\mathbb{E}_\infty$*-space map. This implies* MU *is an* $\mathbb{E}_\infty$*-ring.*

We first want to understand:

$$H_*(MU; \mathbb{Z}) \simeq \pi_*(\mathbb{Z} \otimes_{\mathbb{S}} MU)$$

We look at the diagram:

$$\begin{array}{ccccc} & & \text{Spectra} & \xrightarrow{-\otimes_{\mathbb{S}}\mathbb{Z}} & \mathbb{Z}\text{-mod} \\ & & \uparrow & & \uparrow \\ BU & \longrightarrow & B\mathrm{GL}_1\mathbb{S} & \longrightarrow & B\mathrm{GL}_1(\mathbb{Z}) = \mathbb{RP}^\infty \end{array}$$

But observe that the bottom arrow $BU \to B\mathrm{GL}_1(\mathbb{Z})$ is represented by a class in

$$H^1(BU; \mathbb{F}_2) = 0$$

. So it must be the trivial map. Hence we indeed have:

$$H_*(MU; \mathbb{Z}) \simeq H_*(BU; \mathbb{Z})$$

Remark.

$$\oplus : BU \times BU \to BU$$

makes $H_*(BU; \mathbb{Z})$ *a ring. In fact*

$$H_*(BU; \mathbb{Z}) \simeq \pi_*(\mathbb{Z} \otimes_{\mathbb{S}} MU)$$

is a ring isomorphism.

Note there is a natural map $BU(1) = \mathbb{CP}^\infty \to BU$, we have:

Proposition 10.26.

$$H_*(BU; \mathbb{Z}) \simeq \mathbb{Z}[b_1, b_2, \cdots]$$

is a polynomial ring, where b_i *are the elements in* $\mathrm{im}\,(H_*(\mathbb{CP}^\infty \to H_*BU))$.

Hence:

$$H_*(MU; \mathbb{Z}) \simeq \mathbb{Z}[b_1, b_2, \cdots]; \quad H_*(MU; \mathbb{F}_2) \simeq \mathbb{F}_2[b_1, b_2, \cdots]$$

There is a coaction (this is a ring map):

$$H_*(MU; \mathbb{F}_2) \to H_*(MU; \mathbb{F}_2) \otimes_{\mathbb{F}_2} \mathcal{A}_*$$

Remark. *This coaction* **does not** *agree with that on* $H_*(BU; \mathbb{F}_2)$.

Let's denote F to be the composition:

$$\mathbb{CP}^\infty \to BU \to B\mathrm{GL}_1\mathbb{S} \to \text{Spectra}$$

There is a natural map $\mathrm{colim}(F) \to MU$. But note that:

$$H_*(\mathrm{colim}(F); \mathbb{F}_2) \simeq H_*(\mathbb{CP}^\infty; \mathbb{F}_2) = \mathbb{F}_2\{b_i\}$$

$$H_*(MU; \mathbb{F}_2) \simeq H_*(BU; \mathbb{F}_2) = \mathbb{F}_2[b_i]$$

Under these identifications, the map $H_*(\mathrm{colim}(F); \mathbb{F}_2) \to H_*(MU; \mathbb{F}_2)$ is simply the natural emebdding.

Therefore, to understand the coaction on $H_*(MU; \mathbb{F}_2)$, it suffices to understand the coaction on $H_*(\mathrm{colim}(F); \mathbb{F}_2)$.

Proposition 10.27. $\operatorname{colim}(F) = \Sigma^{-2}\Sigma^\infty \mathbb{CP}^\infty$.

Remark.

$$H_*(\Sigma^{-2}\Sigma^\infty \mathbb{CP}^\infty; \mathbb{F}_2) = H_*(\mathbb{CP}^\infty; \mathbb{F}_2) = H_*(\Sigma^\infty_+ \mathbb{CP}^\infty; \mathbb{F}_2)$$

The Adam Spectral Sequence computes: $\pi_*(\widehat{MU}_2)$ starting with $\operatorname{Ext}_{\mathcal{A}}(H^*(MU; \mathbb{F}_2), \mathbb{F}_2)$. Milnor computed this to have starting page concentrated in degrees such that there are no possible differentials. Something similar can be done for each p. Moreover, we have $\mathbb{Q} \otimes_{\mathbb{S}} MU = \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z} \otimes_{\mathbb{S}} MU$.

Theorem 10.28 (Milnor)**.** *1. $\pi_* MU$ is concentrated in even degrees.*

2. *The map $\pi_*(MU) \to \pi_*(\mathbb{Z} \otimes_{\mathbb{S}} MU) \simeq \mathbb{Z}[b_1, b_2, \cdots]$ is injective.*
3. *The image includes each b_i such that $i+1$ is not a power of prime; if $i+1 = p^k$, then the image includes $p \cdot b_i$+(lower degree terms).*

Milnor's theorem tells us $MU^*\mathbb{CP}^\infty$ is a formal group. Note we have a natural map:

$$\mathbb{CP}^\infty \to BU \to B\mathrm{GL}_1\mathbb{S} \to \text{Spectra}$$

This gives me a natural map:

$$\Sigma^{-2}\Sigma^\infty \mathbb{CP}^\infty \xrightarrow{t} MU$$

In other word $t \in MU^2\mathbb{CP}^\infty$ is a complex orientation. Hence we have a canonical isomorphism

$$MU^*\mathbb{CP}^\infty \simeq \pi_* MU[[t]]$$

realizing $MU^*\mathbb{CP}^\infty$ as a formal group law. More concretely, we have a canonical power series $f(x, y) \in (\pi_* MU)[[x, y]]$ satisfying the axioms of formal group laws.

Definition 10.29. The **Lizard ring** $L \coloneqq \mathbb{Z}[a_{ij}]/$ (relations enforced by a_{ij} being coefficients of a formal group law).

There is a natural map $L \to \pi_* MU$ implementing the coefficients of f.

Remark. *For any discrete ring R, a map $L \to R$ is the same data as a formal group of coefficient in R.*

Theorem 10.30 (Quillen)**.** *The map*

$$L \to \pi_*(MU) \to \pi_*(\mathbb{Z} \otimes_{\mathbb{S}} MU) \simeq \mathbb{Z}[b_i]$$

is injective. Its image contains all b_i such that $i+1$ is not a prime power, and we can find explicit terms $p \cdot b_i + \cdots$ for $i+1$ a p-power.

Corollary 10.31. *$L \to \pi_* MU$ is an isomoprhism. Namely $\pi_* MU$ is a universal formal group.*

Proof. (Of Quillen's Theorem, sketched). The idea is to compare two complex orientations on $\mathbb{Z} \otimes MU$. Note $\mathbb{Z} \otimes MU$ is an even $\mathbb{E}_\infty$-ring, there is an isomorphism:

$$(\mathbb{Z} \otimes MU)^* \mathbb{CP}^\infty \simeq \mathbb{Z}[b_i][[t_{MU}]]$$

for

$$t_{MU} : \Sigma^{-2}\Sigma^\infty \mathbb{CP}^\infty \to MU \xrightarrow{1 \otimes \mathrm{id}} \mathbb{Z} \otimes MU$$

On the other hand $\mathbb{Z}^* \mathbb{CP}^\infty = \mathbb{Z}[t_\mathbb{Z}]$ gives a different complex orientation:

$$\Sigma^{-2}\Sigma^\infty \mathbb{CP}^\infty \xrightarrow{t_\mathbb{Z}} \mathbb{Z} \xrightarrow{\mathrm{id} \otimes 1} \mathbb{Z} \otimes MU$$

We have

$$\mathbb{Z}[b_i][[t_{MU}]] = \mathbb{Z}[b_i][[t_\mathbb{Z}]]$$

These two complex orientations are related by:

Lemma. $t_{MU} = t_\mathbb{Z} + b_1(t_\mathbb{Z})^2 + b_2(t_\mathbb{Z})^3 + \cdots$

But the formal group structure on $(MU \otimes \mathbb{Z})^* \mathbb{CP}^\infty$ gives me a map:

$$(MU \otimes \mathbb{Z})^* \mathbb{CP}^\infty = \mathbb{Z}[b_i][[t_{MU}]] \to (MU \otimes \mathbb{Z})^*(\mathbb{CP}^\infty \times \mathbb{CP}^\infty) = \mathbb{Z}[b_i][[x_{MU}, y_{MU}]]$$

$$t_{MU} \mapsto x_{MU} + y_{MU} + \sum a_{ij} x_{MU}^i y_{MU}^j$$

and

$$t_\mathbb{Z} \mapsto x_\mathbb{Z} + y_\mathbb{Z}$$

This tells me:

$$(x_\mathbb{Z} + y_\mathbb{Z}) + b_1(x_\mathbb{Z} + y_\mathbb{Z})^2 + \cdots = x_{MU} + y_{MU} + \sum a_{ij} x_{MU}^i y_{MU}^j$$

Compare the coefficients give the explicit formula. For example, we can directly see $a_{11} = 2b_1$. □

10.5 More on Formal Group Laws

Given a discrete ring R, a formal group law over R is a power series $f(x, y) \in R[[x, y]]$ satsifying:

1. $f(x, 0) = f(0, x) = x$
2. $f(x, y) = f(y, x)$
3. $f(f(x, y), z) = f(x, f(y, z))$

Recall we proved that formal group laws are in bijection with maps $L = \pi_* MU \to R$.

Remark. *Suppose $g(t) = t + b_1 t^2 + b_2 t^3 + \cdots$ is a power series in $R[[t]]$. Then there exists $g^{-1} = t + \cdots$. Then if $f(x, y)$ is a formal group law, so is $g^{-1}(f(g(x), g(y)))$. We say these two formal group laws are strictly isomorphic.*

Definition 10.32. The moduli stack $\mathcal{M}_{\text{fg}}^{\text{strict}}$ is the functor:

$$\text{Rings} \to \text{Groupoids}$$

$$R \mapsto \{\text{formal group laws over } R \text{ with strict isomorphisms}\}$$

Remark. *If $\lambda \in R^\times$, and $f(x,y)$ is a formal group law, then so is $\lambda f(\lambda^{-1}x, \lambda^{-1}y)$. This can be thought of as an action of $\mathbb{G}_m$ on $\mathcal{M}_{\text{fg}}^{\text{strict}}$, then $\mathcal{M}_{\text{fg}} = \mathcal{M}_{\text{fg}}^{\text{strict}}/\mathbb{G}_m$*

Remark. *Suppose k is algebraically closed field.*

1. *If k has characteristic 0, all formal group laws over k are isomorphic to $f(x,y) = x+y$.*
2. *If k has characteristic p, every formal group law has a height $1 \leq h \leq \infty$. They're classified by their height.*

To determine height, consider (e.g. $p = 5$):

$$f(f(f(f(x,x),x),x),x)$$

is a power series $px + \ell_2 x^2 + \ell_3 x^3 + \cdots$. The formal group law has height $\geq h$ if the coefficient of x^{p^n} (i.e. ℓ_{p^n}) is not a unit.

Note $\pi_*(MU) = L$. We can compute $MU_*MU \simeq \pi_* MU[b_1, b_2, \cdots]$ using the Postnikov filtration of MU. Compare with our previous result $\mathbb{Z}_{**}MU \simeq \mathbb{Z}[b_1, b_2, \cdots]$

To compute $\pi_*\mathbb{S}$, we can do descent using:

$$\mathbb{S} = \mathbb{S}_{MU} = \lim(MU^{\otimes_{\mathbb{S}} \bullet+1})$$

So there is a spectral sequence, starting with the homology of the chain complex:

$$\pi_*(MU) \to \pi_*(MU \otimes_{\mathbb{S}} MU) \to \pi_*(MU \otimes_{\mathbb{S}} MU \otimes_{\mathbb{S}} MU) \to \cdots$$

A map $\pi_* MU \to R$ is a formal group law over R.

A map $\pi_*(MU \otimes_{\mathbb{S}} MU) \to R$ is a pair of formal group laws witha strict isomorphism between them.

A map $\pi_*(MU \otimes_{\mathbb{S}} MU \otimes_{\mathbb{S}} MU) \to R$ is a diagram

$$f_1 \xrightarrow{g_1} f_2 \xrightarrow{g_2} f_3$$

is $\mathcal{M}_{\text{fg}}^{\text{strict}}(R)$.

The upshot: the starting page of this spectral sequence is

$$H^*(\mathcal{M}_{\text{fg}}^{\text{strict}}, \mathcal{O})$$

The $\mathbb{G}_m$-action on $\mathcal{M}_{\text{fg}}^{\text{strict}}$ gives this a bigrading. This is the Adams-Novikov / motivic spectral sequence.

Remark. *If X is any spectrum, there is a coaction*

$$MU_*X \to (MU_*X) \otimes_{\pi_* MU} MU_*MU$$

This will imply MU_*X is a quasi-coherent sheaf over $\mathcal{M}_{\text{fg}}$.

In classical algebraic geometry, $\mathcal{M}_{\text{fg}}$, the closed points are classified by a pair (p, h) for p a prime, h a height. Given a sheaf over $\mathcal{M}_{\text{fg}}$, we can "localize" or "complete".

We have a functor:

$$\text{Spectra} \to \text{Shv}(\mathcal{M}_{\text{fg}})$$

$$X \mapsto MU_*X$$

p-localization / completion of a spectrum focues on one tower of points. $X \mapsto X \otimes_{\mathbb{S}} \mathbb{Q}$ focuses on one point.

For any prime p and height h, there is a functor:

$$L_{K(h)} : \text{Spectra} \to \text{Spectra}$$

which "focues at one point".

Remark. *Let* $\mathbb{S}_{KU}$ *denote* $\lim(KU^{\otimes_{\mathbb{S}} \cdot + 1})$. *Then*

$$L_{K(1)}X \coloneqq (X \widehat{\otimes_{\mathbb{S}} \mathbb{S}_{KU}})_p$$

While $\mathbb{S}$ is complicated, we might hope to entirely understand

$$H^*(\mathcal{M}_{\text{fg}}^{\text{strict}}, \mathcal{O}_{K(n)}) \Rightarrow \pi_*(L_{K(h)}\mathbb{S})$$

for any fixed h. It's done when $h = 1$ and $h = 2$.

Theorem 10.33 (Hopkins-Ravenel)**.** *This spectral sequence has finitely many pages and at the end is concentrated in a band of degrees.*